

Variational differential quadrature: A technique to simplify numerical analysis of structures



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ABSTRACT

A new solution method in the area of computational mechanics is developed in this article, which is called variational differential quadrature (VDQ). The main idea of this method is based on the accurate and direct discretization of the energy functional in the structural mechanics. In the VDQ method, through developing an efficient matrix formulation and using an accurate integral operator, the discretized governing equations are derived directly from the weak form of the equations with no need for the analytical derivation of the strong form. This technique provides an alternative way to discretize the energy functional, which avoids the local interpolation and the assembly process of the methods of this kind. We first implement the VDQ method for the nonlinear elasticity theory considering the Green-St. Venant strain tensor; then we simplify the formulation further for the first-order shear deformable beam and plate theories. The final formulation of these cases demonstrates the simplicity of the implementation for the VDQ method in the numerical analysis of the structures, which is a major goal for this article. Using these examples, one can easily learn and apply this technique to other structures. To assess the performance of the VDQ method, we compare it with the generalized differential quadrature (GDQ) method and finite element method (FEM) in the case of bending analysis of Mindlin plates. It is indicated that computational cost of VDQ is less than that of GDQ, and the convergence rate of VDQ is faster than that of FEM.

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1. Introduction

A large number of the problems arising in engineering can be formulated in terms of energy functionals, i.e. the weak form, from which the governing differential equations of the strong form can be derived. Also, extremization of those functionals is equivalent to the solution of the resulting differential equations. Developing solution methods within both weak and strong frameworks is of great importance to the researchers in computational mechanics. From this point of view, solution techniques can be classified into two main groups. In the first group, there are methods that are applied to the strong forms, such as the finite difference method (FDM) [1–4], the differential quadrature method (DQM) [5–12] and the generalized differential quadrature (GDQ) method [13–17]. In the second group, methods are based on the weak forms and the variational formulations, such as the Rayleigh–Ritz method [18–21], the Galerkin method [22–25], the meshless local Petrov–Galerkin (MLPG) method [26,27] and the finite element method [28]. Among the aforementioned methods, DQM, GDQ, and FEM have been the subject of much attention due to their capability to solve a variety of problems.

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The idea of the Differential Quadrature Method (DQM) was first introduced by Bellman et al. [5,6]. Consider a set of discrete points along a direction in a domain, known as the grid points. DQM approximates the derivative of a function with respect to that direction at any of those discrete points using a weighted linear sum of the function values evaluated at all discrete points. Approximating the derivative at each point needs a different set of weighting coefficients with a size equal to the number of the grid points. Therefore, estimating the derivative over n discrete points needs a weighting coefficient matrix of $n \times n$. For the first-order derivative, Bellman et al. [6] explained two approaches for calculating the weighting coefficient matrix. The first one is solving a set of n algebraic equations, which are obtained by direct evolution of derivatives of n test functions e.g., $1, x, x^2, \dots, x^{n-1}$ on the given grid points. As explained in [8,15], the corresponding matrix of this system of linear equations becomes ill-conditioned as the number of grid points increases, and evaluating the inverse of the matrix becomes increasingly inaccurate. The other approach of Bellman et al. [6] provides an explicit formulation for evaluating the weighting coefficient matrix. However, the relations are valid only by choosing the roots of a shifted Legendre polynomial of degree n as the underlying grid points. This could be restrictive in analysis of different physical problems since the distribution of points is always fixed. These shortcomings are completely overcome by the work of Shu and Richards [13,14], the generalized differential quadrature (GDQ) method. Using the Lagrange polynomials as the test functions, they introduced a simple explicit formula for the weighting coefficients of the first-order derivative, which is valid for any arbitrary grid points. Moreover, Shu and Richards [14] presented an explicit recurrence relation for calculating the weighting coefficients of the higher-order derivatives up to order of $n-1$. These practical formulations of GDQ method are given in the Appendix (see Eq. (a.1)). Interestingly, it can be proved that the weighting coefficients of GDQ are exactly equivalent to the coefficients of the highest-order finite difference scheme [29]. Also, the GDQ method and the Chebyshev collocation method result in identical approximation whenever the grid points are chosen to be the same for both methods [13,15]. The reader is referred to [8] for a comprehensive review on differential quadrature methods and their applications. Moreover, Shu's book [15] is a valuable reference to study the DQ methods thoroughly. Differential quadrature methods possess the advantages of high degree of accuracy and fast convergence. Moreover, using a proper matrix formulation, these methods can be implemented easily especially using MATLAB programming language. However, these techniques are applied to the partial differential equations (PDEs) and, in structural mechanics, the processes of derivation of the governing PDEs from the energy functional may be difficult and tedious especially for non-conventional problems.

Finite element methods can be directly applied to the energy functional, and the process of deriving the governing PDEs can be avoided. However, FEM has some other difficulties in its implementation such as defining proper local shape functions, calculating the affine mappings from the reference element, generating a proper mesh, performing the assembly process and solving possible instability and locking problems in some structures. Therefore, the implementation of FEM may not be straightforward even for standard problems. Also, its convergence rate is slower than that of DQ methods.

As the third approach, a mixed method of differential quadrature and global integral quadrature, which approximates the integral over the entire domain, can be utilized to discretize the weak formulations, leading to an algebraic system of equations. The integral quadrature method approximates the integral of a function by a weighted linear sum of the discrete values of that function. An early idea of global approximation of the integral using the quadrature rule was presented by Civan [30]. More efficiently, the integral over an arbitrary grid points can be approximated globally using the generalized integral quadrature (GIQ) method, which was proposed by Shu [13,15]. Since GDQ and GIQ are both accurate and can be implemented on the same grid points, their combination can be used to discretize the weak equations and integro-differential equations. Raju et al. [31] and White et al. [32,33] effectively implemented this approach in their studies to obtain the discretized equations for postbuckling analysis of the plates and shells.

Prompted by the aforementioned explanations, the aim of this paper is to develop a new solution technique in the area of computational mechanics named as Variational Differential Quadrature (VDQ) method. By introducing a novel matrix formulation and using an accurate integral operator, this method provides the capability of direct discretization of the energy functional using the standard differential quadrature operators. A similar but much restricted approach was previously developed by the authors in [34] based on the Rayleigh–Ritz method for linear vibrational analysis of composite conical shells. The solution procedure in that study could be an example of direct discretization of the energy functional but just for a specific linear problem. The VDQ method is much more general than the one presented in [34]. It is developed and explained in details for three-dimensional nonlinear elasticity. Therefore, it can be learned fully and reduced readily for the analysis of structural engineering components such as beams, plates, and shells. As examples, the nonlinear VDQ formulation for the Timoshenko beam theory and the Mindlin plate theory are also presented in this work.

Simplicity in implementation is a major feature of the VDQ method. As an example, consider the nonlinear analysis of the Mindlin plate. This is a classical and standard problem; however, implementation of the available methods and their corresponding codes for this problem is not completely trivial. In the DQ methods, one needs to derive the governing nonlinear PDEs of the plate using the variational calculus. Even then, the discretization of those PDEs needs considerable effort. In addition, derivation of the Jacobian matrix for Newton–Raphson method from the discretized PDEs is complicated, so instead, it is obtained numerically, which slows the computations and increases the approximation error. Using FEM, derivation of PDEs is excluded, but developing an efficient FEM code to solve the nonlinear problems of Mindlin plate is clearly a challenging task and needs considerable programming skills. These difficulties in implementation are completely resolved by the VDQ method. For example, in the case of the Mindlin plate, the nonlinear discretized governing equations as well as their corresponding Jacobian matrix are obtained by a few compact relations, which are derived directly from the energy functional. Accordingly, the corresponding codes for this problem can be written in just few lines in MATLAB. This claim, i.e.

simplicity in the implementation of the VDQ method, is also valid for a non-standard problem. For instance, the final VDQ formulations for three-dimensional nonlinear elasticity, as they are derived in this paper, are simple, compact, and easy to implement.

At first glance, the content of this paper and especially the process of deriving the VDQ equations may be considered tedious. But, as an introduction to this method, we tried to review needed concepts, include the details of all calculations, and prove all relations and identities. Once one masters the basic idea and the simple approach of the VDQ method, he/she can easily write the final relations intuitively just by looking at the energy functional and the strain tensor. For clarification on this as well as motivation, we recommend the readers to take a look at [Section 7](#), summary of final equations, before going through the paper.

The paper is organized as follows. A general outline for the VDQ method is given in [Section 2](#). In [Section 3.1](#), needed relations of the nonlinear elasticity are reviewed for the homogeneous and isotropic materials in three-dimensional Cartesian coordinates. Strain energy density is written in terms of the Green-St. Venant strain tensor. In [Section 3.2](#), a practical matrix representation is introduced for the relations of [Section 3.1](#). The terms of Hamilton's principle including the strain energy for nonlinear elasticity are discretized using the VDQ method in [Section 4.1](#) and [Section 4.2](#). Accordingly, the discretized VDQ equations of motion for nonlinear elasticity are derived in [Section 4.3](#). Derivation of the tangent stiffness matrix, which is used as the Jacobian matrix in the Newton–Raphson method, is presented in [Section 4.4](#). Properties of the tangent stiffness matrix are discussed in [Section 4.5](#). Implementation of the boundary conditions is explained in [Section 4.6](#). For more insight, the VDQ formulations of the Timoshenko beam theory and the Mindlin plate theory are derived in [Sections 5](#) and [6](#), respectively. [Section 7](#) is dedicated to the summary of all final VDQ formulations derived in [Sections 4, 5, and 6](#). Finally, numerical examples and results are presented in [Section 8](#). Conclusions are given in [Section 9](#). The equations for the differential operators, the integral operators, and the grid points are given in the Appendix. Furthermore, an outline of error analysis for the VDQ method is provided in the Appendix.

2. Outline of the VDQ method

In this section, we explain a general outline for the VDQ method without going to the details.

Suppose $\mathbf{U}$ is a vector that contains some smooth real-valued functions u_i , which are defined on a subset of $\mathbb{R}^n$. Let consider a functional F that can be expressed in the following form

$$F(\mathbf{U}) = \int \left(\frac{1}{2} \mathbf{U}_1^T \mathbf{C} \mathbf{U}_2 - \mathbf{V}^T \mathbf{A} \mathbf{G} + \mathbf{H} \right) d\Omega,$$

where $\mathbf{U}_1$, $\mathbf{U}_2$, and $\mathbf{V}$ are vectors that contain some derivatives of the functions u_i and possibly some of the functions u_i themselves. $\mathbf{V}$ contains linear terms, e.g., $\frac{\partial u_1}{\partial x_2}$ or u_2 , but $\mathbf{U}_1$ and $\mathbf{U}_2$ may contain higher-order terms, e.g., $(\frac{\partial u_1}{\partial x_1})(\frac{\partial u_1}{\partial x_2})$ or $(\frac{\partial u_2}{\partial x_3})^3$. $\mathbf{G}$ and $\mathbf{H}$ are vectors that contain some constants and some functions other than u_i . Also, $\mathbf{C}$ and $\mathbf{A}$ are constant matrices. Note that most energy functionals such as the total strain energy and the total kinetic energy can be written in the form of F assumed above.

Using the VDQ method, one can approximate the extremum of the above functional on some grid points. For simplicity, assume $\mathbf{U}_1$ and $\mathbf{U}_2$ contain only linear terms. The steps can be summarized as follows:

- (1) Writing $\mathbf{U}_1 = \mathbf{E}_1 \mathbf{U}$, $\mathbf{U}_2 = \mathbf{E}_2 \mathbf{U}$, and $\mathbf{V} = \mathbf{E}_3 \mathbf{U}$ such that the matrices $\mathbf{E}_1$, $\mathbf{E}_2$, and $\mathbf{E}_3$ contain the derivative operators of $\mathbf{U}_1$, $\mathbf{U}_2$, and $\mathbf{V}$, respectively, and some constants. e.g., $\frac{\partial u_1}{\partial x_1}$ in $\mathbf{V}$ places the derivative operator $\frac{\partial}{\partial x_1}$ in $\mathbf{E}_3$ and u_1 in $\mathbf{U}$.
- (2) The functional becomes $F = \int (\frac{1}{2} \mathbf{U}^T \mathbf{E}_1^T \mathbf{C} \mathbf{E}_2 \mathbf{U} - \mathbf{U}^T \mathbf{E}_3^T \mathbf{A} \mathbf{G} + \mathbf{H}) d\Omega$.
- (3) Discretizing $\mathbf{U}_1 = \mathbf{E}_1 \mathbf{U}$, $\mathbf{U}_2 = \mathbf{E}_2 \mathbf{U}$, $\mathbf{V} = \mathbf{E}_3 \mathbf{U}$, $\mathbf{G}$ and $\mathbf{H}$ on some grid points to get $\mathbf{U}_1 = \mathbb{E}_1 \mathbf{U}$, $\mathbf{U}_2 = \mathbb{E}_2 \mathbf{U}$, $\mathbf{V} = \mathbb{E}_3 \mathbf{U}$, $\mathbf{G}$ and $\mathbb{H}$. Here $\mathbb{E}_i$ contain the differential matrix operators, e.g., $\frac{\partial}{\partial x_1}$ in $\mathbf{E}_3$ is replaced with the matrix $\mathbf{D}_x^{(1)}$ to form $\mathbb{E}_3$.
- (4) The discretized functional becomes $F = \frac{1}{2} \mathbf{U}^T \mathbb{E}_1^T (\mathbf{C} \otimes \mathbb{S}) \mathbb{E}_2 \mathbf{U} - \mathbf{U}^T \mathbb{E}_3^T (\mathbf{A} \otimes \mathbb{S}) \mathbf{G} + (\mathbf{I} \otimes \mathbb{S}) \mathbb{H}$, where $\mathbb{S}$ is the diagonal matrix of the integral operator, $\otimes$ denotes the Kronecker product, and $\mathbf{I}$ is an identity matrix, for which $\mathbf{H} = \mathbf{I} \mathbf{H}$ holds.
- (5) The variation of the functional is $\delta F = \delta \mathbf{U}^T (\frac{1}{2} \mathbb{E}_1^T (\mathbf{C} \otimes \mathbb{S}) \mathbb{E}_2 \mathbf{U} + \frac{1}{2} \mathbb{E}_2^T (\mathbf{C}^T \otimes \mathbb{S}) \mathbb{E}_1 \mathbf{U} + \mathbb{E}_3^T (\mathbf{A} \otimes \mathbb{S}) \mathbf{G})$.
- (6) Defining $\mathbb{K} = \frac{1}{2} (\mathbb{E}_1^T (\mathbf{C} \otimes \mathbb{S}) \mathbb{E}_2 + \mathbb{E}_2^T (\mathbf{C}^T \otimes \mathbb{S}) \mathbb{E}_1)$ and $\mathbb{F} = \mathbb{E}_3^T (\mathbf{A} \otimes \mathbb{S}) \mathbf{G}$, and setting $\delta F = 0$ to get $\mathbb{K} \mathbf{U} = \mathbb{F}$.
- (7) Imposing the boundary conditions and solving $\mathbb{K} \mathbf{U} = \mathbb{F}$ to find the extremum $\mathbf{U}$.

If $\mathbf{U}_1 = \mathbf{U}_2$ and $\mathbf{V} = \mathbf{U}$, and $\mathbf{C}$ is symmetric, then $\mathbb{K} = \mathbb{E}^T (\mathbf{C} \otimes \mathbb{S}) \mathbb{E}$ and $\mathbb{F} = (\mathbf{A} \otimes \mathbb{S}) \mathbf{G}$.

If $\mathbf{U}_1$ and $\mathbf{U}_2$ contain higher-order terms, steps (2) and (4) remain unchanged, but we have the following modifications: Further considerations are needed in step (1) to formulate $\mathbf{E}_1$ and $\mathbf{E}_2$ in terms of $\mathbf{U}$ properly (e.g. see [Section 3.2](#)). Accordingly, in step (3), $\mathbb{E}_1$ and $\mathbb{E}_2$ become some functions of $\mathbf{U}$ (e.g. see [Section 4.2](#)). In step (5), variations of $\mathbb{E}_1(\mathbf{U})$ and $\mathbb{E}_2(\mathbf{U})$ must be taken. In step (6), $\mathbb{K}$ becomes a function of $\mathbf{U}$. We solve a nonlinear equation $\mathbb{K}(\mathbf{U}) \mathbf{U} = \mathbb{F}$ using the Newton method in step (7).

We will show the validity of step (4) in [Section 4.1](#) and [Section 4.2](#). The differential matrix operators, the integral operators, and the grid points relations are discussed in the Appendix.

In the following sections, we will go carefully through the above steps while we are applying the VDQ method to the nonlinear elasticity, Timoshenko beam, and Mindlin plate theories. To have a better perspective on the VDQ method, see the summary of final equations in [Section 7](#).

3. Nonlinear elasticity theory

In this work, we use the St. Venant–Kirchhoff material model, which uses the constitutive equation of the linear elastic materials (Hooke's law), but is written in terms of the nonlinear Green–St. Venant strain tensor.

3.1. Tensorial representation

The Green–St. Venant strain tensor is given by

$$\begin{aligned}\tilde{\mathbf{E}} &= \frac{1}{2}(\nabla(\mathbf{U}) + \nabla^T(\mathbf{U}) + \nabla^T(\mathbf{U})\nabla(\mathbf{U})), \\ \tilde{\mathcal{E}}_{ij} &= \frac{1}{2}(u_{i,j} + u_{j,i} + u_{k,i}u_{k,j}) = \tilde{\mathcal{E}}_{ji}.\end{aligned}\quad (3.1)$$

One can rewrite Eq. (3.1) as

$$\begin{aligned}\tilde{\mathcal{E}}_{11} &= \frac{\partial u_1}{\partial x_1} + \frac{1}{2}\left(\left(\frac{\partial u_1}{\partial x_1}\right)^2 + \left(\frac{\partial u_2}{\partial x_1}\right)^2 + \left(\frac{\partial u_3}{\partial x_1}\right)^2\right), \\ \tilde{\mathcal{E}}_{22} &= \frac{\partial u_2}{\partial x_2} + \frac{1}{2}\left(\left(\frac{\partial u_1}{\partial x_2}\right)^2 + \left(\frac{\partial u_2}{\partial x_2}\right)^2 + \left(\frac{\partial u_3}{\partial x_2}\right)^2\right), \\ \tilde{\mathcal{E}}_{33} &= \frac{\partial u_3}{\partial x_3} + \frac{1}{2}\left(\left(\frac{\partial u_1}{\partial x_3}\right)^2 + \left(\frac{\partial u_2}{\partial x_3}\right)^2 + \left(\frac{\partial u_3}{\partial x_3}\right)^2\right), \\ \tilde{\mathcal{E}}_{23} &= \tilde{\mathcal{E}}_{32} = \frac{1}{2}\left(\frac{\partial u_2}{\partial x_3} + \frac{\partial u_3}{\partial x_2} + \frac{\partial u_1}{\partial x_2}\frac{\partial u_1}{\partial x_3} + \frac{\partial u_2}{\partial x_2}\frac{\partial u_2}{\partial x_3} + \frac{\partial u_3}{\partial x_2}\frac{\partial u_3}{\partial x_3}\right), \\ \tilde{\mathcal{E}}_{13} &= \tilde{\mathcal{E}}_{31} = \frac{1}{2}\left(\frac{\partial u_1}{\partial x_3} + \frac{\partial u_3}{\partial x_1} + \frac{\partial u_1}{\partial x_1}\frac{\partial u_1}{\partial x_3} + \frac{\partial u_2}{\partial x_1}\frac{\partial u_2}{\partial x_3} + \frac{\partial u_3}{\partial x_1}\frac{\partial u_3}{\partial x_3}\right), \\ \tilde{\mathcal{E}}_{12} &= \tilde{\mathcal{E}}_{21} = \frac{1}{2}\left(\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} + \frac{\partial u_1}{\partial x_1}\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1}\frac{\partial u_2}{\partial x_2} + \frac{\partial u_3}{\partial x_1}\frac{\partial u_3}{\partial x_2}\right).\end{aligned}\quad (3.2)$$

The strain energy density is a function of strain. For homogeneous and isotropic materials, the strain energy density is expressed as

$$\hat{U}(\tilde{\mathbf{E}}) = \frac{1}{2}\lambda\tilde{\mathcal{E}}_{ii}\tilde{\mathcal{E}}_{jj} + \mu\tilde{\mathcal{E}}_{ij}\tilde{\mathcal{E}}_{ij} + \sigma(|\tilde{\mathbf{E}}|^2), \quad (3.3)$$

where λ and μ are the Lamé parameters. It is assumed that the strains are small enough so that one can neglect the higher-order terms in Eq. (3.3). The strain energy density is thus given by

$$\hat{U}(\tilde{\mathbf{E}}) = \frac{1}{2}\lambda\tilde{\mathcal{E}}_{ii}\tilde{\mathcal{E}}_{jj} + \mu\tilde{\mathcal{E}}_{ij}\tilde{\mathcal{E}}_{ij}. \quad (3.4)$$

Materials that obey this law are named St. Venant–Kirchhoff materials. This material model is the simplest one among the nonlinear elasticity models, because the constitutive relation is linear and only the strain tensor is nonlinear. This model is widely used in the computational mechanics and engineering structures.

Let $\tilde{\mathbf{S}}$ be the second Piola–Kirchhoff stress tensor. One has

$$\tilde{\mathcal{S}}_{ij} = \frac{\partial \hat{U}(\tilde{\mathbf{E}})}{\partial \tilde{\mathcal{E}}_{ij}}. \quad (3.5)$$

By means of Eqs. (3.4) and (3.5), one can obtain the components of $\tilde{\mathbf{S}}$ as follows

$$\tilde{\mathcal{S}}_{ij} = \lambda\mathcal{E}_{kk}\delta_{ij} + 2\mu\mathcal{E}_{ij}. \quad (3.6)$$

According to the linear relation between the second Piola–Kirchhoff stress tensor and the Green–St. Venant strain tensor, one can write the above equation as

$$\tilde{\mathbf{S}} = \tilde{\mathbf{C}} : \tilde{\mathbf{E}}, \quad (3.7)$$

where the fourth-order tensor $\tilde{\mathbf{C}}$ is called the elasticity tensor. According to Eq. (3.6), the components of the elasticity tensor are derived as

$$\tilde{\mathcal{C}}_{ijkl} = \lambda\delta_{ij}\delta_{kl} + 2\mu\delta_{ik}\delta_{jl}. \quad (3.8)$$

Now, using Eqs. (3.4) and (3.6), one can obtain the following relation

$$\hat{U} = \frac{1}{2} \tilde{\mathcal{E}} : \tilde{\mathcal{S}} = \frac{1}{2} \tilde{\mathcal{E}}_{ij} \tilde{\mathcal{S}}_{ij}, \quad (3.9)$$

which in accordance with Eq. (3.7) can be written as

$$\hat{U} = \frac{1}{2} \tilde{\mathcal{E}} : \tilde{\mathcal{C}} : \tilde{\mathcal{E}} = \frac{1}{2} \tilde{\mathcal{E}}_{ij} \tilde{\mathcal{C}}_{ijkl} \tilde{\mathcal{E}}_{kl}. \quad (3.10)$$

3.2. Matrix representation

From a computational point of view, the matrix form of relations is preferable to their tensorial form. Thus, in this section, we express the strain energy using a matrix relation. To this end, vectorization and matricization processes should be carried out on the strain, stress, and elasticity tensors.

The Green-St. Venant strain vector can be written as

$$\mathcal{E} = [\tilde{\mathcal{E}}_{11} \quad \tilde{\mathcal{E}}_{22} \quad \tilde{\mathcal{E}}_{33} \quad 2\tilde{\mathcal{E}}_{23} \quad 2\tilde{\mathcal{E}}_{13} \quad 2\tilde{\mathcal{E}}_{12}]^T, \quad (3.11)$$

which can be expressed in terms of displacement vector as follows

$$\mathcal{E} = \mathbf{E}\mathbf{U} + \sum_{i=1}^3 (\mathbf{G}_{2i-1}\mathbf{U}) \circ (\mathbf{G}_{2i}\mathbf{U}), \quad (3.12)$$

where $\circ$ denotes the Hadamard product. Also, the displacement vector is

$$\mathbf{U} = [u_1 \quad u_2 \quad u_3]^T. \quad (3.13)$$

Moreover, the matrix $\mathbf{E}$ is formulated as

$$\mathbf{E} = \begin{bmatrix} \partial/\partial x_1 & 0 & 0 \\ 0 & \partial/\partial x_2 & 0 \\ 0 & 0 & \partial/\partial x_3 \\ 0 & \partial/\partial x_3 & \partial/\partial x_2 \\ \partial/\partial x_3 & 0 & \partial/\partial x_1 \\ \partial/\partial x_2 & \partial/\partial x_1 & 0 \end{bmatrix}. \quad (3.14)$$

The term $\mathbf{E}\mathbf{U}$ in Eq. (3.12) gives the linear part of strain; by defining six matrices $\mathbf{G}_1$ to $\mathbf{G}_6$ and using the Hadamard product, one can construct the nonlinear part of strain. In this regard, the row vector $\mathbf{e}_i$ and the column vectors $\mathbf{A}_1$ and $\mathbf{A}_2$ are defined as follows

$$\mathbf{e}_i = [\delta_{i1} \quad \delta_{i2} \quad \delta_{i3}], \quad (3.15)$$

$$\mathbf{A}_1 = \begin{bmatrix} \frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_2} & \frac{\partial}{\partial x_3} & \frac{\partial}{\partial x_2} & \frac{\partial}{\partial x_1} & \frac{\partial}{\partial x_1} \end{bmatrix}^T, \quad (3.16)$$

$$\mathbf{A}_2 = \begin{bmatrix} \frac{1}{2} \frac{\partial}{\partial x_1} & \frac{1}{2} \frac{\partial}{\partial x_2} & \frac{1}{2} \frac{\partial}{\partial x_3} & \frac{\partial}{\partial x_3} & \frac{\partial}{\partial x_3} & \frac{\partial}{\partial x_2} \end{bmatrix}^T, \quad (3.17)$$

where in Eq. (3.15) δ_{ij} stands for the Kronecker delta which is given by

$$\delta_{ij} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases} \quad (3.18)$$

Now, it is possible to define the operators $\mathbf{G}_1$ to $\mathbf{G}_6$ as

$$\mathbf{G}_{2i-1} = \mathbf{e}_i \otimes \mathbf{A}_1, \quad \mathbf{G}_{2i} = \mathbf{e}_i \otimes \mathbf{A}_2, \quad i = 1, 2, 3. \quad (3.19)$$

Let $\langle \mathbf{X} \rangle := \text{diag}(\mathbf{X})$ be the diagonal matrix obtained from the elements of the vector $\mathbf{X}$. The following relation exists for the Hadamard product of two vectors $\mathbf{X}$ and $\mathbf{Y}$

$$\mathbf{X} \circ \mathbf{Y} = \langle \mathbf{X} \rangle \mathbf{Y} = \langle \mathbf{Y} \rangle \mathbf{X} = \frac{1}{2} (\langle \mathbf{X} \rangle \mathbf{Y} + \langle \mathbf{Y} \rangle \mathbf{X}). \quad (3.20)$$

Hence

$$\begin{aligned} (\mathbf{G}_{2i-1}\mathbf{U}) \circ (\mathbf{G}_{2i}\mathbf{U}) &= \langle \mathbf{G}_{2i-1}\mathbf{U} \rangle \mathbf{G}_{2i}\mathbf{U} = \langle \mathbf{G}_{2i}\mathbf{U} \rangle \mathbf{G}_{2i-1}\mathbf{U} \\ &= \frac{1}{2} (\langle \mathbf{G}_{2i-1}\mathbf{U} \rangle \mathbf{G}_{2i} + \langle \mathbf{G}_{2i}\mathbf{U} \rangle \mathbf{G}_{2i-1})\mathbf{U} = \frac{1}{2} \mathbf{E}_{n;i}\mathbf{U}, \quad i = 1, 2, 3, \end{aligned} \quad (3.21)$$

in which $\mathbf{E}_{n;i}$ is defined as follows

$$\mathbf{E}_{n;i} = \langle \mathbf{G}_{2i-1}\mathbf{U} \rangle \mathbf{G}_{2i} + \langle \mathbf{G}_{2i}\mathbf{U} \rangle \mathbf{G}_{2i-1}, \quad i = 1, 2, 3. \quad (3.22)$$

Accordingly, one can write Eq. (3.12) as

$$\boldsymbol{\varepsilon} = \mathbf{E}\mathbf{U} + \frac{1}{2} \sum_{i=1}^3 \mathbf{E}_{n,i} \mathbf{U}. \quad (3.23)$$

Now, by defining $\mathbf{E}_n$ in the following form

$$\mathbf{E}_n = \sum_{i=1}^3 \mathbf{E}_{n,i}, \quad (3.24)$$

which is a first-order relation with respect to $\mathbf{U}$, one can express the Green-St. Venant strain vector as

$$\boldsymbol{\varepsilon} = \left(\mathbf{E} + \frac{1}{2} \mathbf{E}_n \right) \mathbf{U}. \quad (3.25)$$

In fact, in this relation, $\mathbf{E}$ and $\frac{1}{2} \mathbf{E}_n$ are the linear and the nonlinear strain operators, respectively. It is worth mentioning that formulating the nonlinear parts of strain using Eq. (3.21) leads to symmetry of the stiffness matrices in the final equations.

Now, if the second Piola–Kirchhoff stress tensor is vectorized as

$$\mathbf{S} = [\tilde{S}_{11} \quad \tilde{S}_{22} \quad \tilde{S}_{33} \quad \tilde{S}_{23} \quad \tilde{S}_{13} \quad \tilde{S}_{12}]^T, \quad (3.26)$$

and the fourth-order elasticity tensor is matricized as

$$\mathbf{C} = \begin{bmatrix} \lambda + 2\mu & \lambda & \lambda & 0 & 0 & 0 \\ & \lambda + 2\mu & \lambda & 0 & 0 & 0 \\ & & \lambda + 2\mu & 0 & 0 & 0 \\ & & & \mu & 0 & 0 \\ & & & & \mu & 0 \\ \text{sym.} & & & & & \mu \end{bmatrix}, \quad (3.27)$$

then Eq. (3–7) can be represented in the following matrix form

$$\mathbf{S} = \mathbf{C}\boldsymbol{\varepsilon}. \quad (3.28)$$

Now, by the obtained relations, one can represent the strain energy density in a matrix form as

$$\hat{U} = \frac{1}{2} \boldsymbol{\varepsilon} \cdot \mathbf{S} = \frac{1}{2} \boldsymbol{\varepsilon}^T \mathbf{S} = \frac{1}{2} \boldsymbol{\varepsilon}^T \mathbf{C}\boldsymbol{\varepsilon} = \frac{1}{2} \varepsilon_i C_{ij} \varepsilon_j. \quad (3.29)$$

The total strain energy in the region Ω is thus written as

$$U = \int_{\Omega} \hat{U} dV = \frac{1}{2} \int_{\Omega} \boldsymbol{\varepsilon}^T \mathbf{C}\boldsymbol{\varepsilon} dV = \frac{1}{2} \int_{\Omega} \varepsilon_i C_{ij} \varepsilon_j dV. \quad (3.30)$$

4. VDQ formulation for the nonlinear elasticity theory

4.1. Discretization of Hamilton's principle

Let T , U , and W_{ext} respectively denote the kinetic energy, the strain energy, and the work done by external forces. Hamilton's principle is formulated as

$$\int_{t_1}^{t_2} (\delta T - \delta U + \delta W_{ext}) dt = 0, \quad (4.1)$$

in which T , U and W_{ext} in the region Ω can be written as

$$T = \frac{1}{2} \int_{\Omega} \dot{\mathbf{U}}^T \rho \dot{\mathbf{U}} dV = \frac{1}{2} \int_{\Omega} \dot{u}_i \rho \dot{u}_i dV, \quad (4.2)$$

$$W_{ext} = \int_{\Omega} \mathbf{U}^T \tilde{\mathbf{F}} dV = \int_{\Omega} u_i \tilde{F}_i dV, \quad (4.3)$$

$$U = \int_{\Omega} \hat{U} dV = \frac{1}{2} \int_{\Omega} \varepsilon_i C_{ij} \varepsilon_j dV, \quad (4.4)$$

where ρ and $\tilde{\mathbf{F}}$ show the mass density and the vector of external forces, respectively.

It is possible to discretize the integrands of Eqs. (4.2)–(4.4) by considering n points in the region Ω , and then, to obtain the energy values using an integral operator (see Section b of the Appendix). By calculating the component u_i of displacement vector at every point, one can construct the following vector

$$\mathbf{U}_I = [u_i(x_1) \quad u_i(x_2) \quad \cdots \quad u_i(x_n)]^T. \quad (4.5)$$

Hence, the discretized displacement vector becomes

$$\mathbb{U} = \begin{bmatrix} \mathbf{U}_1 \\ \mathbf{U}_2 \\ \mathbf{U}_3 \end{bmatrix}. \quad (4.6)$$

The force vector and strain energy density are also constructed on the discrete points as follows

$$\tilde{\mathbf{F}} = \begin{bmatrix} \tilde{\mathbf{F}}_1 \\ \tilde{\mathbf{F}}_2 \\ \tilde{\mathbf{F}}_3 \end{bmatrix}, \quad (4.7)$$

$$\hat{\mathbf{U}} = [\hat{U}(x_1) \quad \hat{U}(x_2) \quad \dots \quad \hat{U}(x_n)]^T. \quad (4.8)$$

Using the discretized vectors of displacement and force, one can write Eqs. (4.2)–(4.4) as

$$T = \frac{1}{2} \int_{\Omega} \rho \dot{\mathbf{U}}_I \circ \dot{\mathbf{U}}_I \, dV, \quad (4.9)$$

$$W_{ext} = \int_{\Omega} \mathbf{U}_I \circ \tilde{\mathbf{F}}_I \, dV, \quad (4.10)$$

$$U = \int_{\Omega} \hat{\mathbf{U}} dV. \quad (4.11)$$

Now, the integrals of Eqs. (4.2)–(4.4) can be evaluated on the discretized region by defining a suitable integral operator (see Section b of the Appendix). Let $\tilde{\mathbf{S}}$ be a vector with the size of $1 \times n$. Then, one has

$$T = \frac{1}{2} \rho \tilde{\mathbf{S}} (\dot{\mathbf{U}}_I \circ \dot{\mathbf{U}}_I), \quad (4.12)$$

$$W_{ext} = \tilde{\mathbf{S}} (\mathbf{U}_I \circ \tilde{\mathbf{F}}_I), \quad (4.13)$$

$$U = \tilde{\mathbf{S}} \hat{\mathbf{U}}. \quad (4.14)$$

For converting the vector $\tilde{\mathbf{S}}$ into a diagonal matrix, one can write

$$\mathbf{S} = \text{diag}(\tilde{\mathbf{S}}) =: \langle \tilde{\mathbf{S}} \rangle. \quad (4.15)$$

Thus, via the following relation for some column vectors $\mathbf{V}_1$, $\mathbf{V}_2$ and $\mathbf{V}_3$

$$\mathbf{V}_1^T (\mathbf{V}_2 \circ \mathbf{V}_3) = \mathbf{V}_2^T \langle \mathbf{V}_1 \rangle \mathbf{V}_3, \quad (4.16)$$

Eqs. (4.12) and (4.13) are written as

$$T = \frac{1}{2} \dot{\mathbf{U}}_I^T \rho \mathbf{S} \dot{\mathbf{U}}_I = \frac{1}{2} \dot{\mathbf{U}}_I^T \rho \delta_{ij} \mathbf{S} \dot{\mathbf{U}}_J, \quad (4.17)$$

$$W_{ext} = \mathbf{U}_I^T \mathbf{S} \tilde{\mathbf{F}}_I = \mathbf{U}_I^T \delta_{ij} \mathbf{S} \tilde{\mathbf{F}}_J. \quad (4.18)$$

It should be noted that in the preceding equations $i=I$ and $j=J$, where lowercase indices are for scalars and uppercase indices belong to vectors. The Kronecker product of matrices $\mathbf{A}$ and $\mathbf{B}$ is expressed as

$$\mathbb{P} = \mathbf{A} \otimes \mathbf{B} = \begin{bmatrix} a_{11} \mathbf{B} & a_{12} \mathbf{B} & \dots \\ a_{21} \mathbf{B} & a_{22} \mathbf{B} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}. \quad (4.19)$$

The matrix $a_{ij} \mathbf{B}$ can be represented by $\mathbf{P}_{IJ}$ in which $i=I$ and $j=J$. Accordingly, one has

$$\mathbf{P}_{IJ} = (\mathbf{A} \otimes \mathbf{B})_{IJ} = a_{ij} \mathbf{B}, \quad (4.20)$$

$$\mathbb{P} = \begin{bmatrix} \mathbf{P}_{11} & \mathbf{P}_{12} & \dots \\ \mathbf{P}_{21} & \mathbf{P}_{22} & \dots \\ \vdots & \vdots & \ddots \end{bmatrix}. \quad (4.21)$$

Using Eqs. (4.20) and (4.21), one can write Eqs. (4.17) and (4.18) as follows

$$T = \frac{1}{2} \dot{\mathbf{U}}_I^T \rho \delta_{ij} \mathbf{S} \dot{\mathbf{U}}_J = \frac{1}{2} \dot{\mathbf{U}}_I^T (\rho \mathbf{I}_3 \otimes \mathbf{S})_{IJ} \dot{\mathbf{U}}_J = \frac{1}{2} \dot{\mathbf{U}}^T (\rho \mathbf{I}_3 \otimes \mathbf{S}) \dot{\mathbf{U}}, \quad (4.22)$$

$$W_{ext} = \mathbf{U}_I^T \delta_{ij} \mathbf{S} \tilde{\mathbf{F}}_J = \mathbf{U}_I^T (\mathbf{I}_3 \otimes \mathbf{S})_{IJ} \tilde{\mathbf{F}}_J = \mathbf{U}^T (\mathbf{I}_3 \otimes \mathbf{S}) \tilde{\mathbf{F}}, \quad (4.23)$$

where $\mathbf{I}_3$ is a 3×3 identity matrix. By defining the following matrices

$$\mathbb{M} = \rho \mathbf{I}_3 \otimes \mathbf{S}, \quad (4.24)$$

$$\mathbb{F} = (\mathbf{I}_3 \otimes \mathbb{S}) \tilde{\mathbb{F}}, \quad (4.25)$$

one has

$$T = \frac{1}{2} \dot{\mathbf{U}}^T \mathbb{M} \dot{\mathbf{U}}, \quad (4.26)$$

$$W_{ext} = \mathbf{U}^T \mathbb{F}. \quad (4.27)$$

Based on the following relation

$$(\mathbf{A} \otimes \mathbf{B})^T = \mathbf{A}^T \otimes \mathbf{B}^T \quad (4.28)$$

and bringing to mind that the identity matrix $\mathbf{I}_3$ and the diagonal matrix $\mathbb{S}$ are symmetric, one concludes that the matrix $\mathbb{M}$ is symmetric.

At this step, using the derived equations, one can obtain δT and δW_{ext} as follows

$$\delta T = \frac{1}{2} \left(\delta \dot{\mathbf{U}}^T \mathbb{M} \dot{\mathbf{U}} + \dot{\mathbf{U}}^T \mathbb{M} \delta \dot{\mathbf{U}} \right), \quad (4.29)$$

$$\delta W_{ext} = \delta \mathbf{U}^T \mathbb{F}. \quad (4.30)$$

By introducing the following relation

$$\mathbf{V}_1^T \mathbf{A} \mathbf{V}_2 = \mathbf{V}_2^T \mathbf{A}^T \mathbf{V}_1 \quad (4.31)$$

and owing to the symmetry of matrix $\mathbb{M}$, one can write

$$\dot{\mathbf{U}}^T \mathbb{M} \delta \dot{\mathbf{U}} = \delta \dot{\mathbf{U}}^T \mathbb{M} \dot{\mathbf{U}}. \quad (4.32)$$

Therefore, Eq. (4.29) is converted to

$$\delta T = \delta \dot{\mathbf{U}}^T \mathbb{M} \dot{\mathbf{U}}. \quad (4.33)$$

Based upon Eqs. (4.30) and (4.33), Hamilton's principle becomes

$$\int_{t_1}^{t_2} \left(\delta \dot{\mathbf{U}}^T \mathbb{M} \dot{\mathbf{U}} - \delta U + \delta \mathbf{U}^T \mathbb{F} \right) dt. \quad (4.34)$$

Integrating by parts results in

$$\begin{aligned} \int_{t_1}^{t_2} \delta \dot{\mathbf{U}}^T \mathbb{M} \dot{\mathbf{U}} dt &= \delta \mathbf{U}^T \mathbb{M} \dot{\mathbf{U}} \Big|_{t_1}^{t_2} - \int_{t_1}^{t_2} \delta \mathbf{U}^T \mathbb{M} \ddot{\mathbf{U}} dt \\ &= \delta \mathbf{U}^T(t_2) \mathbb{M} \dot{\mathbf{U}}(t_2) - \delta \mathbf{U}^T(t_1) \mathbb{M} \dot{\mathbf{U}}(t_1) - \int_{t_1}^{t_2} \delta \mathbf{U}^T \mathbb{M} \ddot{\mathbf{U}} dt. \end{aligned} \quad (4.35)$$

Since Hamilton's principle is written for a time interval between two moments with definite status, the variation of displacement vector in those statuses is equal to zero. Hence, $\delta \mathbf{U}^T(t_2) = \delta \mathbf{U}^T(t_1) = 0$, which leads to

$$\delta T = - \int_{t_1}^{t_2} \delta \mathbf{U}^T \mathbb{M} \ddot{\mathbf{U}} dt. \quad (4.36)$$

Accordingly, the discretized form of Hamilton's principle can be expressed as

$$\int_{t_1}^{t_2} \left(-\delta \mathbf{U}^T \mathbb{M} \ddot{\mathbf{U}} - \delta U + \delta \mathbf{U}^T \mathbb{F} \right) dt = 0. \quad (4.37)$$

In the next section, the calculation of δU is discussed.

4.2. Discretization of strain energy

To discretize the strain energy, it is first necessary to discretize the strain vector. Let $\mathcal{E}_i$ denote the discretized form of the components $\mathcal{E}_i$ of the strain vector given by Eq. (3.11). Then, the discretized strain vector can be represented by the following column vector

$$\bar{\mathcal{E}} = \left[\mathcal{E}_1^T \quad \mathcal{E}_2^T \quad \mathcal{E}_3^T \quad \mathcal{E}_4^T \quad \mathcal{E}_5^T \quad \mathcal{E}_6^T \right]^T. \quad (4.38)$$

Now, one can write the discretized form of Eq. (3.25) as follows

$$\bar{\mathcal{E}} = \left(\mathbb{E} + \frac{1}{2} \mathbb{E}_n \right) \mathbf{U}, \quad (4.39)$$

where $\mathbf{U}$ is the discretized displacement vector. The operator $\mathbb{E}$ is obtained by

$$\mathbb{E} = \begin{bmatrix} \mathbf{I}_{n_3} \otimes \mathbf{I}_{n_2} \otimes \mathbf{D}_{x_1}^{(1)} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{n_3} \otimes \mathbf{D}_{x_2}^{(1)} \otimes \mathbf{I}_{n_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{D}_{x_3}^{(1)} \otimes \mathbf{I}_{n_2} \otimes \mathbf{I}_{n_1} \\ \mathbf{0} & \mathbf{D}_{x_3}^{(1)} \otimes \mathbf{I}_{n_2} \otimes \mathbf{I}_{n_1} & \mathbf{I}_{n_3} \otimes \mathbf{D}_{x_2}^{(1)} \otimes \mathbf{I}_{n_1} \\ \mathbf{D}_{x_3}^{(1)} \otimes \mathbf{I}_{n_2} \otimes \mathbf{I}_{n_1} & \mathbf{0} & \mathbf{I}_{n_3} \otimes \mathbf{I}_{n_2} \otimes \mathbf{D}_{x_1}^{(1)} \\ \mathbf{I}_{n_3} \otimes \mathbf{D}_{x_2}^{(1)} \otimes \mathbf{I}_{n_1} & \mathbf{I}_{n_3} \otimes \mathbf{I}_{n_2} \otimes \mathbf{D}_{x_1}^{(1)} & \mathbf{0} \end{bmatrix}, \quad (4.40)$$

where n_i is the number of grid points along the x_i direction, $\mathbf{D}_{x_i}^{(1)}$ is the GDQ differential operator along the x_i direction (see the Appendix), and $\mathbf{I}_{n_i}$ shows a $n_i \times n_i$ identity matrix. Also, the operator $\mathbb{E}_n$ is calculated as follows

$$\mathbb{E}_n = \sum_{i=1}^3 \mathbb{E}_{n;i}, \quad (4.41)$$

$$\mathbb{E}_{n;i} = \langle \mathbb{G}_{2i-1} \mathbf{U} \rangle \mathbb{G}_{2i} + \langle \mathbb{G}_{2i} \mathbf{U} \rangle \mathbb{G}_{2i-1}, \quad i = 1, 2, 3, \quad (4.42)$$

$$\mathbb{G}_{2i-1} = \mathbf{e}_i \otimes \mathbb{A}_1, \quad i = 1, 2, 3, \quad (4.43)$$

$$\mathbb{G}_{2i} = \mathbf{e}_i \otimes \mathbb{A}_2, \quad i = 1, 2, 3, \quad (4.44)$$

$$\mathbb{A}_1 = \begin{bmatrix} \mathbf{I}_{n_3} \otimes \mathbf{I}_{n_2} \otimes \mathbf{D}_{x_1}^{(1)} \\ \mathbf{I}_{n_3} \otimes \mathbf{D}_{x_2}^{(1)} \otimes \mathbf{I}_{n_1} \\ \mathbf{D}_{x_3}^{(1)} \otimes \mathbf{I}_{n_2} \otimes \mathbf{I}_{n_1} \\ \mathbf{I}_{n_3} \otimes \mathbf{D}_{x_2}^{(1)} \otimes \mathbf{I}_{n_1} \\ \mathbf{I}_{n_3} \otimes \mathbf{I}_{n_2} \otimes \mathbf{D}_{x_1}^{(1)} \\ \mathbf{I}_{n_3} \otimes \mathbf{I}_{n_2} \otimes \mathbf{D}_{x_1}^{(1)} \end{bmatrix}, \quad (4.45)$$

$$\mathbb{A}_2 = \begin{bmatrix} \frac{1}{2} \mathbf{I}_{n_3} \otimes \mathbf{I}_{n_2} \otimes \mathbf{D}_{x_1}^{(1)} \\ \frac{1}{2} \mathbf{I}_{n_3} \otimes \mathbf{D}_{x_2}^{(1)} \otimes \mathbf{I}_{n_1} \\ \frac{1}{2} \mathbf{D}_{x_3}^{(1)} \otimes \mathbf{I}_{n_2} \otimes \mathbf{I}_{n_1} \\ \mathbf{D}_{x_3}^{(1)} \otimes \mathbf{I}_{n_2} \otimes \mathbf{I}_{n_1} \\ \mathbf{D}_{x_3}^{(1)} \otimes \mathbf{I}_{n_2} \otimes \mathbf{I}_{n_1} \\ \mathbf{I}_{n_3} \otimes \mathbf{D}_{x_2}^{(1)} \otimes \mathbf{I}_{n_1} \end{bmatrix}. \quad (4.46)$$

By calculating the discretized strain vector, one can calculate the strain energy density $\hat{U}$ at all grid points and then organize them in a column vector shown by $\hat{\mathbf{U}}$. Finally, the total strain energy can be calculated through numerical integration of $\hat{\mathbf{U}}$. Therefore, according to Eq. (3.29), $\hat{\mathbf{U}}$ becomes

$$\hat{\mathbf{U}} = \frac{1}{2} C_{ij} \mathcal{E}_i \circ \mathcal{E}_j. \quad (4.47)$$

Note that in the previous equation, $i=l$ and $j=j$. Based on Eq. (4.14), one has

$$U = \frac{1}{2} C_{ij} \tilde{\mathcal{S}}(\mathcal{E}_i \circ \mathcal{E}_j). \quad (4.48)$$

Using the relation $\mathbf{V}_1^T(\mathbf{V}_2 \circ \mathbf{V}_3) = \mathbf{V}_2^T(\mathbf{V}_1) \mathbf{V}_3$ and by considering that $\mathbb{S} = \langle \tilde{\mathcal{S}} \rangle$, one can obtain the strain energy by

$$U = \frac{1}{2} \mathcal{E}_i^T C_{ij} \mathbb{S} \mathcal{E}_j. \quad (4.49)$$

Also, using the relation $A_{ij} \mathbf{B} = (\mathbf{A} \otimes \mathbf{B})_{ij}$, one can write

$$U = \frac{1}{2} \mathcal{E}_i^T (\mathbf{C} \otimes \mathbb{S})_{ij} \mathcal{E}_j = \frac{1}{2} \bar{\mathcal{E}}^T (\mathbf{C} \otimes \mathbb{S}) \bar{\mathcal{E}}. \quad (4.50)$$

Substituting Eq. (4.39) into the above equation gives

$$U = \frac{1}{2} \mathbf{U}^T \left(\mathbb{E}^T + \frac{1}{2} \mathbb{E}_n^T \right) (\mathbf{C} \otimes \mathbb{S}) \left(\mathbb{E} + \frac{1}{2} \mathbb{E}_n \right) \mathbf{U}. \quad (4.51)$$

The variation of U is required for the discretized Hamilton's principle given by Eq. (4.37). Thus,

$$\delta U = \delta \left(\mathbf{U}^T \left(\mathbb{E}^T + \frac{1}{2} \mathbb{E}_n^T \right) \right) (\mathbf{C} \otimes \mathbb{S}) \left(\mathbb{E} + \frac{1}{2} \mathbb{E}_n \right) \mathbf{U}. \quad (4.52)$$

Due to the symmetry of the elasticity matrix $\mathbf{C}$ and the diagonal matrix $\mathbb{S}$, the matrix $\mathbf{C} \otimes \mathbb{S}$ is symmetric. Hence, consistent with $\mathbf{V}_1^T \mathbf{A} \mathbf{V}_2 = \mathbf{V}_2^T \mathbf{A}^T \mathbf{V}_1$, the following equation holds

$$\delta \left(\mathbf{U}^T \left(\mathbb{E}^T + \frac{1}{2} \mathbb{E}_n^T \right) \right) (\mathbf{C} \otimes \mathbb{S}) \left(\mathbb{E} + \frac{1}{2} \mathbb{E}_n \right) \mathbf{U} = \mathbf{U}^T \left(\mathbb{E}^T + \frac{1}{2} \mathbb{E}_n^T \right) (\mathbf{C} \otimes \mathbb{S}) \delta \left(\left(\mathbb{E} + \frac{1}{2} \mathbb{E}_n \right) \mathbf{U} \right). \quad (4.53)$$

For calculating $\delta(\mathbf{U}^T(\mathbb{E}^T + \frac{1}{2}\mathbb{E}_n^T))$, it should be noted that $\mathbb{E}$ is an expression with the order of zero with respect to $\mathbf{U}$, whereas $\mathbb{E}_n$ is a first-order expression with respect to $\mathbf{U}$. Thus,

$$\delta \left(\mathbf{U}^T \left(\mathbb{E}^T + \frac{1}{2} \mathbb{E}_n^T \right) \right) = \delta \mathbf{U}^T \mathbb{E}^T + \frac{1}{2} \delta \mathbf{U}^T \mathbb{E}_n^T + \frac{1}{2} \mathbf{U}^T \delta \mathbb{E}_n^T. \quad (4.54)$$

Now, using Eqs. (4.41) and (4.42) results in

$$\mathbf{U}^T \delta \mathbb{E}_n^T = \sum_{i=1}^3 \mathbf{U}^T \delta \mathbb{E}_{n,i}^T = \sum_{i=1}^3 \left(\mathbf{U}^T \mathbf{G}_{2i}^T \langle \mathbf{G}_{2i-1} \delta \mathbf{U} \rangle + \mathbf{U}^T \mathbf{G}_{2i-1}^T \langle \mathbf{G}_{2i} \delta \mathbf{U} \rangle \right). \quad (4.55)$$

In addition, according to Eq. (3.20), the following equation holds

$$\mathbf{U}^T \delta \mathbb{E}_{n,i}^T = \delta \mathbf{U}^T \mathbf{G}_{2i-1}^T \langle \mathbf{G}_{2i} \mathbf{U} \rangle + \delta \mathbf{U}^T \mathbf{G}_{2i}^T \langle \mathbf{G}_{2i-1} \mathbf{U} \rangle = \delta \mathbf{U}^T \mathbb{E}_{n,i}^T. \quad (4.56)$$

Therefore,

$$\mathbf{U}^T \delta \mathbb{E}_n^T = \delta \mathbf{U}^T \mathbb{E}_n^T. \quad (4.57)$$

As a result, Eq. (4.54) can be written as

$$\delta \left(\mathbf{U}^T \left(\mathbb{E}^T + \frac{1}{2} \mathbb{E}_n^T \right) \right) = \delta \mathbf{U}^T \mathbb{E}^T + \delta \mathbf{U}^T \mathbb{E}_n^T. \quad (4.58)$$

Finally, the variation of strain energy becomes

$$\delta U = \delta \mathbf{U}^T \left(\mathbb{E}^T + \mathbb{E}_n^T \right) (\mathbf{C} \otimes \mathbb{S}) \left(\mathbb{E} + \frac{1}{2} \mathbb{E}_n \right) \mathbf{U}. \quad (4.59)$$

Now, by introducing the following matrix

$$\tilde{\mathbb{K}}_s = \mathbb{K} + \frac{1}{2} \tilde{\mathbb{N}}_1 + \frac{1}{3} \mathbb{N}_2, \quad (4.60)$$

where

$$\mathbb{K} = \mathbb{E}^T (\mathbf{C} \otimes \mathbb{S}) \mathbb{E}, \quad (4.61)$$

$$\tilde{\mathbb{N}}_1 = \mathbb{E}^T (\mathbf{C} \otimes \mathbb{S}) \mathbb{E}_n + 2 \mathbb{E}_n^T (\mathbf{C} \otimes \mathbb{S}) \mathbb{E}, \quad (4.62)$$

$$\mathbb{N}_2 = \frac{3}{2} \mathbb{E}_n^T (\mathbf{C} \otimes \mathbb{S}) \mathbb{E}_n, \quad (4.63)$$

one can write the variation of strain energy as follows

$$\delta U = \delta \mathbf{U}^T \tilde{\mathbb{K}}_s \mathbf{U}. \quad (4.64)$$

4.3. Discretized equation of motion

Inserting Eq. (4.64) into (4.37) leads to

$$\int_{t_1}^{t_2} \delta \mathbf{U}^T (-\mathbf{M} \ddot{\mathbf{U}} - \tilde{\mathbb{K}}_s \mathbf{U} + \mathbf{F}) dt = 0. \quad (4.65)$$

Using the fundamental lemma of calculus of variation, one obtains the equation of motion as follows

$$\mathbf{M} \ddot{\mathbf{U}} + \tilde{\mathbb{K}}_s \mathbf{U} - \mathbf{F} = \mathbf{0}, \quad (4.66)$$

where $\tilde{\mathbb{K}}_s$ given by Eq. (4.60) is a non-symmetric matrix, which is not suitable from a computational point of view. Hence, $\tilde{\mathbb{K}}_s$ should be symmetrized. In Eq. (4.60), matrices $\mathbb{K}$ and $\mathbb{N}_2$ are symmetric, but $\tilde{\mathbb{N}}_1$ is non-symmetric. To symmetrize $\tilde{\mathbb{N}}_1$, one can write Eq. (4.62) as follows

$$\tilde{\mathbb{N}}_1 = \mathbb{E}^T (\mathbf{C} \otimes \mathbb{S}) \mathbb{E}_n + \mathbb{E}_n^T (\mathbf{C} \otimes \mathbb{S}) \mathbb{E} + \mathbb{E}_n^T (\mathbf{C} \otimes \mathbb{S}) \mathbb{E}. \quad (4.67)$$

In this equation, the first two terms construct a symmetric expression, because

$$(\mathbb{E}^T (\mathbf{C} \otimes \mathbb{S}) \mathbb{E}_n + \mathbb{E}_n^T (\mathbf{C} \otimes \mathbb{S}) \mathbb{E})^T = \mathbb{E}_n^T (\mathbf{C} \otimes \mathbb{S}) \mathbb{E} + \mathbb{E}^T (\mathbf{C} \otimes \mathbb{S}) \mathbb{E}_n. \quad (4.68)$$

Thus, it is enough to symmetrize the last term of Eq. (4.67). To accomplish this aim, one writes

$$\tilde{\mathbf{N}}_1 \mathbf{U} = (\mathbf{E}^T(\mathbf{C} \otimes \mathbf{S})\mathbf{E}_n + \mathbf{E}_n^T(\mathbf{C} \otimes \mathbf{S})\mathbf{E})\mathbf{U} + \mathbf{E}_n^T(\mathbf{C} \otimes \mathbf{S})\mathbf{E}\mathbf{U}. \quad (4.69)$$

According to Eq. (4.41), the last term of preceding equation can be written as

$$\mathbf{E}_n^T(\mathbf{C} \otimes \mathbf{S})\mathbf{E}\mathbf{U} = (\mathbf{E}_{n;1}^T + \mathbf{E}_{n;2}^T + \mathbf{E}_{n;3}^T)(\mathbf{C} \otimes \mathbf{S})\mathbf{E}\mathbf{U}. \quad (4.70)$$

Moreover, based on Eq. (4.42), one has

$$\mathbf{E}_{n;i}^T(\mathbf{C} \otimes \mathbf{S})\mathbf{E}\mathbf{U} = (\mathbf{G}_{2i-1}^T \langle \mathbf{G}_{2i} \mathbf{U} \rangle + \mathbf{G}_{2i}^T \langle \mathbf{G}_{2i-1} \mathbf{U} \rangle)(\mathbf{C} \otimes \mathbf{S})\mathbf{E}\mathbf{U}. \quad (4.71)$$

Finally, using Eq. (3.20) results in

$$\mathbf{E}_{n;i}^T(\mathbf{C} \otimes \mathbf{S})\mathbf{E}\mathbf{U} = (\mathbf{G}_{2i-1}^T \langle (\mathbf{C} \otimes \mathbf{S})\mathbf{E}\mathbf{U} \rangle \mathbf{G}_{2i} + \mathbf{G}_{2i}^T \langle (\mathbf{C} \otimes \mathbf{S})\mathbf{E}\mathbf{U} \rangle \mathbf{G}_{2i-1})\mathbf{U}. \quad (4.72)$$

By introducing the following matrix

$$\mathbf{K}_1 = \sum_{i=1}^3 (\mathbf{G}_{2i-1}^T \langle (\mathbf{C} \otimes \mathbf{S})\mathbf{E}\mathbf{U} \rangle \mathbf{G}_{2i} + \mathbf{G}_{2i}^T \langle (\mathbf{C} \otimes \mathbf{S})\mathbf{E}\mathbf{U} \rangle \mathbf{G}_{2i-1}), \quad (4.73)$$

one can write Eq. (4.70) as

$$\mathbf{E}_n^T(\mathbf{C} \otimes \mathbf{S})\mathbf{E}\mathbf{U} = \mathbf{K}_1 \mathbf{U}. \quad (4.74)$$

Substituting this equation into Eq. (4.69) gives

$$\tilde{\mathbf{N}}_1 \mathbf{U} = (\mathbf{E}^T(\mathbf{C} \otimes \mathbf{S})\mathbf{E}_n + \mathbf{E}_n^T(\mathbf{C} \otimes \mathbf{S})\mathbf{E} + \mathbf{K}_1)\mathbf{U} = \mathbf{N}_1 \mathbf{U}, \quad (4.75)$$

in which the symmetric matrix $\mathbf{N}_1$ is defined as

$$\mathbf{N}_1 = \mathbf{E}^T(\mathbf{C} \otimes \mathbf{S})\mathbf{E}_n + \mathbf{E}_n^T(\mathbf{C} \otimes \mathbf{S})\mathbf{E} + \mathbf{K}_1. \quad (4.76)$$

Consequently, the symmetric matrix $\mathbf{K}_s$, which is called the secant stiffness matrix, is defined as

$$\mathbf{K}_s = \mathbf{K} + \frac{1}{2}\mathbf{N}_1 + \frac{1}{3}\mathbf{N}_2. \quad (4.77)$$

Matrices $\mathbf{K}$ and $\mathbf{N}_2$ are given by Eqs. (4.61) and (4.63). It should be remarked that matrices $\mathbf{K}$, $\mathbf{N}_1$, and $\mathbf{N}_2$ have the orders of zero, one, and two in regard to $\mathbf{U}$, respectively. Finally, Eq. (4.66), which is the discretized equation of motion, can be written as

$$\mathbf{M}\ddot{\mathbf{U}} + \mathbf{K}_s \mathbf{U} - \mathbf{F} = \mathbf{0}. \quad (4.78)$$

By solving the above equation using the Newton–Raphson method, one can calculate the displacement vector at the grid points.

4.4. Solution of equation of motion

When $\ddot{\mathbf{U}} = \mathbf{0}$, the equilibrium equation becomes

$$\mathbf{K}_s \mathbf{U} - \mathbf{F} = \mathbf{0}. \quad (4.79)$$

So as to solve the above equation via the Newton–Raphson method, one must write the following equations

$$\mathbf{R}(\mathbf{U}) = \mathbf{K}_s \mathbf{U} - \mathbf{F} \Rightarrow \mathbf{R}(\mathbf{U}) = \mathbf{0}. \quad (4.80)$$

Accordingly, the Newton's recursive formula becomes

$$\mathbf{U}_{i+1} = \mathbf{U}_i - \mathbf{K}_t^{-1}(\mathbf{U}_i)\mathbf{R}(\mathbf{U}_i), \quad (4.81)$$

where the Jacobian matrix $\mathbf{K}_t$ is obtained by

$$\mathbf{K}_t(\mathbf{U}) = \frac{d(\mathbf{R}(\mathbf{U}))}{d\mathbf{U}} = \frac{d(\mathbf{K}_s \mathbf{U})}{d\mathbf{U}} = \mathbf{K} + \mathbf{N}_1 + \mathbf{N}_2. \quad (4.82)$$

$\mathbf{K}_t$ is also known as the tangent stiffness matrix.

Herein we prove that $\mathbf{K}_t = \mathbf{K} + \mathbf{N}_1 + \mathbf{N}_2$ holds. In order to calculate Eq. (4.82), the following equation can be utilized

$$d(\mathbf{K}_s \mathbf{U}) = \mathbf{K}_s d\mathbf{U} + d\mathbf{K}_s \mathbf{U}. \quad (4.83)$$

As it can be seen, it is only sufficient to calculate $d\mathbf{K}_s \mathbf{U}$. Knowing that $\mathbf{K}_s = \mathbf{K} + \frac{1}{2}\mathbf{N}_1 + \frac{1}{3}\mathbf{N}_2$, one can calculate the differentiation of each term individually. As $\mathbf{K}$ is not a function of $\mathbf{U}$, one has $d\mathbf{K} = \mathbf{0}$. However, $\mathbf{N}_1$ and $\mathbf{N}_2$ are functions of $\mathbf{U}$, so $d\mathbf{N}_1$ and $d\mathbf{N}_2$ must be calculated. $\mathbf{N}_1$ and $\mathbf{N}_2$ are formulated as

$$\mathbf{N}_1 = \mathbf{E}^T(\mathbf{C} \otimes \mathbf{S})\mathbf{E}_n + \mathbf{E}_n^T(\mathbf{C} \otimes \mathbf{S})\mathbf{E} + \sum_{i=1}^3 (\mathbf{G}_{2i-1}^T \langle (\mathbf{C} \otimes \mathbf{S})\mathbf{E}\mathbf{U} \rangle \mathbf{G}_{2i} + \mathbf{G}_{2i}^T \langle (\mathbf{C} \otimes \mathbf{S})\mathbf{E}\mathbf{U} \rangle \mathbf{G}_{2i-1}), \quad (4.84)$$

$$\mathbf{N}_2 = \frac{3}{2} \mathbf{E}_n^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{E}_n. \quad (4.85)$$

Since in the relations of $\mathbf{N}_1$ and $\mathbf{N}_2$, $\mathbf{E}$ is not a function of $\mathbf{U}$, but $\mathbf{E}_n$ which is given by

$$\mathbf{E}_n = \sum_{i=1}^3 (\langle \mathbf{G}_{2i-1} \mathbf{U} \rangle \mathbf{G}_{2i} + \langle \mathbf{G}_{2i} \mathbf{U} \rangle \mathbf{G}_{2i-1}), \quad (4.86)$$

is a function of $\mathbf{U}$, $d\mathbf{N}_1 \mathbf{U}$ and $d\mathbf{N}_2 \mathbf{U}$ can be written as

$$d\mathbf{N}_1 \mathbf{U} = \mathbf{E}^T (\mathbf{C} \otimes \mathbf{S}) d\mathbf{E}_n \mathbf{U} + d\mathbf{E}_n^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{E} \mathbf{U} + \sum_{i=1}^3 (\mathbf{G}_{2i-1}^T \langle (\mathbf{C} \otimes \mathbf{S}) \mathbf{E} d\mathbf{U} \rangle \mathbf{G}_{2i} + \mathbf{G}_{2i}^T \langle (\mathbf{C} \otimes \mathbf{S}) \mathbf{E} d\mathbf{U} \rangle \mathbf{G}_{2i-1}) \mathbf{U}, \quad (4.87)$$

$$d\mathbf{N}_2 \mathbf{U} = \frac{3}{2} (d\mathbf{E}_n^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{E}_n \mathbf{U} + \mathbf{E}_n^T (\mathbf{C} \otimes \mathbf{S}) d\mathbf{E}_n \mathbf{U}), \quad (4.88)$$

where

$$d\mathbf{E}_n = \sum_{i=1}^3 (\langle \mathbf{G}_{2i-1} d\mathbf{U} \rangle \mathbf{G}_{2i} + \langle \mathbf{G}_{2i} d\mathbf{U} \rangle \mathbf{G}_{2i-1}). \quad (4.89)$$

Therefore, Eq. (4.87) can be expressed as

$$\begin{aligned} d\mathbf{N}_1 \mathbf{U} &= \mathbf{E}^T (\mathbf{C} \otimes \mathbf{S}) \sum_{i=1}^3 (\langle \mathbf{G}_{2i-1} d\mathbf{U} \rangle \mathbf{G}_{2i} \mathbf{U} + \langle \mathbf{G}_{2i} d\mathbf{U} \rangle \mathbf{G}_{2i-1} \mathbf{U}) \\ &\quad + \sum_{i=1}^3 (\mathbf{G}_{2i}^T \langle \mathbf{G}_{2i-1} d\mathbf{U} \rangle (\mathbf{C} \otimes \mathbf{S}) \mathbf{E} \mathbf{U} + \mathbf{G}_{2i-1}^T \langle \mathbf{G}_{2i} d\mathbf{U} \rangle (\mathbf{C} \otimes \mathbf{S}) \mathbf{E} \mathbf{U}) \\ &\quad + \sum_{i=1}^3 (\mathbf{G}_{2i-1}^T \langle (\mathbf{C} \otimes \mathbf{S}) \mathbf{E} d\mathbf{U} \rangle \mathbf{G}_{2i} \mathbf{U} + \mathbf{G}_{2i}^T \langle (\mathbf{C} \otimes \mathbf{S}) \mathbf{E} d\mathbf{U} \rangle \mathbf{G}_{2i-1} \mathbf{U}). \end{aligned} \quad (4.90)$$

Making use of $\mathbf{V}_1 \circ \mathbf{V}_2 = \langle \mathbf{V}_1 \rangle \mathbf{V}_2 = \langle \mathbf{V}_2 \rangle \mathbf{V}_1$, the previous relation is written as

$$\begin{aligned} d\mathbf{N}_1 \mathbf{U} &= \mathbf{E}^T (\mathbf{C} \otimes \mathbf{S}) \sum_{i=1}^3 (\langle \mathbf{G}_{2i} \mathbf{U} \rangle \mathbf{G}_{2i-1} d\mathbf{U} + \langle \mathbf{G}_{2i-1} \mathbf{U} \rangle \mathbf{G}_{2i} d\mathbf{U}) \\ &\quad + \sum_{i=1}^3 (\mathbf{G}_{2i}^T \langle (\mathbf{C} \otimes \mathbf{S}) \mathbf{E} \mathbf{U} \rangle \mathbf{G}_{2i-1} d\mathbf{U} + \mathbf{G}_{2i-1}^T \langle (\mathbf{C} \otimes \mathbf{S}) \mathbf{E} \mathbf{U} \rangle \mathbf{G}_{2i} d\mathbf{U}) \\ &\quad + \sum_{i=1}^3 (\mathbf{G}_{2i-1}^T \langle \mathbf{G}_{2i} \mathbf{U} \rangle (\mathbf{C} \otimes \mathbf{S}) \mathbf{E} d\mathbf{U} + \mathbf{G}_{2i}^T \langle \mathbf{G}_{2i-1} \mathbf{U} \rangle (\mathbf{C} \otimes \mathbf{S}) \mathbf{E} d\mathbf{U}) \\ &= \mathbf{E}^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{E}_n d\mathbf{U} + \sum_{i=1}^3 (\mathbf{G}_{2i}^T \langle (\mathbf{C} \otimes \mathbf{S}) \mathbf{E} \mathbf{U} \rangle \mathbf{G}_{2i-1} + \mathbf{G}_{2i-1}^T \langle (\mathbf{C} \otimes \mathbf{S}) \mathbf{E} \mathbf{U} \rangle \mathbf{G}_{2i}) d\mathbf{U} + \mathbf{E}_n^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{E} d\mathbf{U}, \end{aligned}$$

which results in

$$d\mathbf{N}_1 \mathbf{U} = \mathbf{N}_1 d\mathbf{U}. \quad (4.91)$$

Likewise, Eq. (4.88) is rewritten as

$$d\mathbf{N}_2 \mathbf{U} = \frac{3}{2} \left(d\mathbf{E}_n^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{E}_n \mathbf{U} + \mathbf{E}_n^T (\mathbf{C} \otimes \mathbf{S}) \sum_{i=1}^3 (\langle \mathbf{G}_{2i-1} d\mathbf{U} \rangle \mathbf{G}_{2i} \mathbf{U} + \langle \mathbf{G}_{2i} d\mathbf{U} \rangle \mathbf{G}_{2i-1} \mathbf{U}) \right). \quad (4.92)$$

Using the relation $\mathbf{V}_1 \circ \mathbf{V}_2 = \langle \mathbf{V}_1 \rangle \mathbf{V}_2 = \langle \mathbf{V}_2 \rangle \mathbf{V}_1$, one can simplify the second term of the above equation as follows

$$\begin{aligned} \mathbf{E}_n^T (\mathbf{C} \otimes \mathbf{S}) \sum_{i=1}^3 (\langle \mathbf{G}_{2i-1} d\mathbf{U} \rangle \mathbf{G}_{2i} \mathbf{U} + \langle \mathbf{G}_{2i} d\mathbf{U} \rangle \mathbf{G}_{2i-1} \mathbf{U}) &= \mathbf{E}_n^T (\mathbf{C} \otimes \mathbf{S}) \sum_{i=1}^3 (\langle \mathbf{G}_{2i} \mathbf{U} \rangle \mathbf{G}_{2i-1} d\mathbf{U} + \langle \mathbf{G}_{2i-1} \mathbf{U} \rangle \mathbf{G}_{2i} d\mathbf{U}) \\ &= \mathbf{E}_n^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{E}_n d\mathbf{U} = \frac{2}{3} \mathbf{N}_2 d\mathbf{U}. \end{aligned} \quad (4.93)$$

So, Eq. (4.92) becomes

$$d\mathbf{N}_2 \mathbf{U} = \frac{3}{2} d\mathbf{E}_n^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{E}_n \mathbf{U} + \mathbf{N}_2 d\mathbf{U}. \quad (4.94)$$

Multiplying both sides of this equation by $\mathbf{U}^T$ results in

$$\mathbf{U}^T d\mathbf{N}_2 \mathbf{U} = \frac{3}{2} \mathbf{U}^T d\mathbf{E}_n^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{E}_n \mathbf{U} + \mathbf{U}^T \mathbf{N}_2 d\mathbf{U}. \quad (4.95)$$

Based on the first term of Eq. (4.92) and following a same procedure which was used for Eq. (4.93), one can derive the following equation

$$\mathbf{U}^T d\mathbf{E}_n^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{E}_n \mathbf{U} = d\mathbf{U}^T \mathbf{E}_n^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{E}_n \mathbf{U} = \frac{2}{3} d\mathbf{U}^T \mathbf{N}_2 \mathbf{U}. \quad (4.96)$$

For two vectors $\mathbf{V}_1$ and $\mathbf{V}_2$ and the matrix $\mathbf{M}$, one has $\mathbf{V}_1^T \mathbf{M} \mathbf{V}_2 = \mathbf{V}_2^T \mathbf{M}^T \mathbf{V}_1$. Thus, using the symmetry property of matrix $\mathbf{N}_2$, one can write

$$d\mathbf{U}^T \mathbf{N}_2 \mathbf{U} = \mathbf{U}^T \mathbf{N}_2 d\mathbf{U}. \quad (4.97)$$

Therefore, Eq. (4.95) can be written as

$$\mathbf{U}^T d\mathbf{N}_2 \mathbf{U} = 2\mathbf{U}^T \mathbf{N}_2 d\mathbf{U} \Rightarrow \mathbf{U}^T (d\mathbf{N}_2 \mathbf{U} - 2\mathbf{N}_2 d\mathbf{U}) = 0. \quad (4.98)$$

As $\mathbf{U}$ is a non-zero vector in this equation, one must have

$$d\mathbf{N}_2 \mathbf{U} - 2\mathbf{N}_2 d\mathbf{U} = \mathbf{0} \Rightarrow d\mathbf{N}_2 \mathbf{U} = 2\mathbf{N}_2 d\mathbf{U}. \quad (4.99)$$

Hence

$$d\mathbb{K}_s \mathbf{U} = \frac{1}{2} d\mathbf{N}_1 \mathbf{U} + \frac{1}{3} d\mathbf{N}_2 \mathbf{U} = \left(\frac{1}{2} \mathbf{N}_1 + \frac{2}{3} \mathbf{N}_2 \right) d\mathbf{U}. \quad (4.100)$$

Eq. (4.83) is thus rewritten as

$$\begin{aligned} d(\mathbb{K}_s \mathbf{U}) &= \mathbb{K}_s d\mathbf{U} + d\mathbb{K}_s \mathbf{U} = \left(\mathbb{K} + \frac{1}{2} \mathbf{N}_1 + \frac{1}{3} \mathbf{N}_2 \right) d\mathbf{U} + \left(\frac{1}{2} \mathbf{N}_1 + \frac{2}{3} \mathbf{N}_2 \right) d\mathbf{U} \\ &= (\mathbb{K} + \mathbf{N}_1 + \mathbf{N}_2) d\mathbf{U}. \end{aligned} \quad (4.101)$$

Therefore,

$$\mathbb{K}_t(\mathbf{U}) = \frac{d(\mathbb{K}_s \mathbf{U})}{d\mathbf{U}} = \mathbb{K} + \mathbf{N}_1 + \mathbf{N}_2, \quad (4.102)$$

which shows the validity of Eq. (4.82).

4.5. Properties of tangent stiffness matrix

At each iteration of Newton's method, let say i , we solve the linear system $\mathbb{K}_t(\mathbf{U}_i) \Delta \mathbf{U}_i = \mathbf{F}$, where $\Delta \mathbf{U}_i = \mathbf{U}_{i+1} - \mathbf{U}_i$, thus properties of the tangent stiffness matrix $\mathbb{K}_t$ are of interest from a computational point of view.

In the VDQ method, $\mathbb{K}_t$ is symmetric. As we demonstrated in Section 4.4, $\mathbb{K}_t(\mathbf{U}) = \mathbb{K} + \mathbf{N}_1(\mathbf{U}) + \mathbf{N}_2(\mathbf{U})$, where $\mathbb{K} = \mathbf{E}^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{E}$, and $\mathbf{N}_1$ and $\mathbf{N}_2$ are given by Eqs. (4.84) and (4.85). Since $\mathbf{S}$ is a diagonal matrix, the elasticity matrix $\mathbf{C}$ given by Eq. (3.27) is symmetric, and the Kronecker product of two symmetric matrices results in a symmetric matrix (because $(\mathbf{A} \otimes \mathbf{B})^T = \mathbf{A}^T \otimes \mathbf{B}^T$), we conclude that $\mathbb{K}$ is symmetric. Likewise, $\mathbf{N}_2$ is symmetric. And we obtained $\mathbf{N}_1$ to be the symmetric form of $\tilde{\mathbf{N}}_1$ (see Eqs. (4.67)–(4.76)).

In a linear analysis, $\mathbf{N}_1(\mathbf{U}) = \mathbf{N}_2(\mathbf{U}) = \mathbf{0}$ and $\mathbb{K}_t = \mathbb{K} = \mathbf{E}^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{E}$. For common materials ($0 < E$ and $0 < \nu < 0.5$), the symmetric matrix $\mathbf{C}$ has real positive diagonal elements and is a strictly diagonally dominant matrix, thus it is positive-definite. Also, it is easy to check numerically that all the diagonal entries of the diagonal matrix $\mathbf{S}$ are real and positive when we use Chebyshev grids to obtain the integral operators (see Eqs. (a.7), (a.8), and (b.9) in the Appendix). Moreover, if eigenvalues of $\mathbf{A}$ are a_i , $i=1, 2, \dots, n$ and eigenvalues of $\mathbf{B}$ are b_j , $j=1, 2, \dots, m$, then eigenvalues of $(\mathbf{A} \otimes \mathbf{B})$ are $a_i b_j$ for $i=1, 2, \dots, n$ and $j=1, 2, \dots, m$. Therefore, we conclude that $(\mathbf{C} \otimes \mathbf{S})$ is positive-definite, that is $\mathbf{Y}^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{Y} > \mathbf{0}$ for every non-zero vector $\mathbf{Y}$ of real numbers. Also, if the discretization of the strain tensor is correct, then the matrix $\mathbf{E}$ is full rank after imposing the boundary conditions (removing some columns of $\mathbf{E}$). Thus $\mathbf{E} \mathbf{U} \neq \mathbf{0}$ for every non-zero vector $\mathbf{U}$. Therefore, $\mathbf{U}^T \mathbb{K} \mathbf{U} = \mathbf{U}^T \mathbf{E}^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{E} \mathbf{U} = \mathbf{V}^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{V} > 0$ holds for every non-zero vector $\mathbf{U}$, that is $\mathbb{K} = \mathbf{E}^T (\mathbf{C} \otimes \mathbf{S}) \mathbf{E}$ is positive-definite.

Since $\mathbf{N}_1(\mathbf{U})$ and $\mathbf{N}_2(\mathbf{U})$ are functions of $\mathbf{U}$, it is difficult to study properties of $\mathbb{K}_t(\mathbf{U})$ in nonlinear analyses. However, due to nature of the problem, we expect $\mathbb{K}_t$ to be positive-definite for specific values of $\mathbf{U}$. The principle of minimum potential energy states that for a structure subjected to conservative external forces, among all displacements that satisfy geometric (essential) boundary conditions, those corresponding to the equilibrium state locally extremize the total potential energy, and if the extremum is minimum, the equilibrium is stable. The total potential energy is $\pi = U - W_{ext}$. We demonstrated in previous sections that $\delta\pi = 0$ results in equilibrium equation $\mathbb{K}_s(\mathbf{U}) \mathbf{U} - \mathbf{F} = \mathbf{0}$. Moreover, based on our calculations in Sections 4.1–4.4 and using Taylor series, we have

$$\pi(\mathbf{U}) = \pi(\mathbf{U}^*) + \Delta \mathbf{U}^T (\mathbb{K}_s(\mathbf{U}^*) \mathbf{U}^* - \mathbf{F}) + \frac{1}{2} \Delta \mathbf{U}^T \mathbb{K}_t(\mathbf{U}^*) \Delta \mathbf{U} + R(\Delta \mathbf{U}),$$

where $\Delta \mathbf{U} = \mathbf{U} - \mathbf{U}^*$ and $R(\Delta \mathbf{U})$ is the reminder of the Taylor series. Note that $\mathbb{K}_s(\mathbf{U})\mathbf{U} - \mathbf{F}$ and $\mathbb{K}_t(\mathbf{U})$ are the gradient vector and the Hessian matrix for $\pi(\mathbf{U})$, respectively. In optimization, $\mathbf{U}^*$ is a local minimum point of $\pi(\mathbf{U})$, if $\mathbb{K}_s(\mathbf{U}^*)\mathbf{U}^* - \mathbf{F} = \mathbf{0}$ (necessary condition) and $\mathbb{K}_t(\mathbf{U}^*)$ is positive-definite (sufficient condition) or at least positive semidefinite (second-order necessary condition). Suppose $\{\mathbf{U}_i\}_{i=0}^{\infty}$ is the sequence of Newton iterations for solving $\mathbb{K}_s(\mathbf{U})\mathbf{U} - \mathbf{F} = \mathbf{0}$ that converges to a local minimum point $\mathbf{U}^*$. Actually, for a given precision, we stop at some $\mathbf{U}_k$ and accept it as an approximation of $\mathbf{U}^*$. However, we expect that continuing the Newton iterations makes progress; thus the singularity of $\mathbb{K}_t(\mathbf{U}_k)$ is out of the question, that is $\mathbb{K}_t(\mathbf{U}_k)$ must not be semidefinite. Also if $\mathbb{K}_t(\mathbf{U}_k)$ be negative-definite or indefinite, then $\mathbf{U}_k$ is not a local minimum and could not be an accurate approximation of $\mathbf{U}^*$. We can repeat the same argument for $\mathbf{U}_{k+1}$. Therefore, we can only declare the validity of the solution for those cases that the tangent stiffness matrix $\mathbb{K}_t$ is positive-definite at least in a small neighborhood of $\mathbf{U}^*$. Now if the Newton iterations start with a guess in that neighborhood, then we can use a linear solver at each iteration, which takes the advantage of positive-definiteness and symmetry of $\mathbb{K}_t$. In practice, if the linear solver fails, that is $\mathbb{K}_t$ becomes other than positive-definite, the initial guess must be improved. And if modifying the initial guess does not work, as we explained, no conclusion on the validity of minimization can be made even if a different linear solver works. In problems with small amplitude of the external forces, we use the solution of the linear analysis $\mathbb{K}\mathbf{U} = \mathbf{F}$ as the initial guess for the Newton method. For highly nonlinear problems, we can use a load control procedure, which increases the amplitude of the force incrementally up to the given value and using the solution of the previous stage as the initial guess of the current stage.

In this work, a symmetric positive-definite solver is used to obtain the numerical results of the VDQ method.

4.6. Implementation of boundary conditions

Various techniques can be used to impose the boundary conditions in the VDQ method. The elimination approach, which is used commonly in FEM, can be implemented in the VDQ method in case of imposing the essential boundary conditions. In the elimination procedure, the corresponding columns and rows of the fixed degrees of freedom at the boundary points are removed from the secant stiffness matrix $\mathbb{K}_s$ and the tangent stiffness matrix $\mathbb{K}_t$; accordingly, the force vector $\mathbf{F}$ or the residual vector $\mathbf{R}$ are modified. Implementing the elimination approach in the VDQ method reduces the size of the unknown vector and the stiffness matrices and preserves symmetry and positive definiteness of $\mathbb{K}_t$, which ultimately results in a faster numerical solution.

For imposing the natural boundary conditions and the nonlinear boundary conditions in the VDQ method, one can implement techniques that are used in the solution of partial differential equations using the finite difference method and the differential quadrature method. A detailed and comparison study of these techniques is discussed in [35]. These techniques modify the matrices $\mathbb{K}_s$ and $\mathbb{K}_t$ without preserving their symmetry or positive definiteness.

In the VDQ analysis of structures, the imposition of the essential boundary conditions is sufficient to solve the governing equations, because the natural boundary conditions are automatically satisfied by the variational formulation of the problem. Therefore, unless considering natural or nonlinear boundary conditions is necessary, making use of the elimination approach in the VDQ method is recommended as it does not change the properties of the stiffness matrices.

5. VDQ formulation for the Timoshenko beam theory

According to the Timoshenko beam theory, the displacement field of the beam is given by

$$\mathbf{U}(x_1, x_3) = \begin{bmatrix} u_1(x_1, x_3) \\ u_3(x_1) \end{bmatrix} = \begin{bmatrix} u(x_1) + x_3 \psi(x_1) \\ w(x_1) \end{bmatrix}, \quad (5.1)$$

where x_1 and x_3 are the coordinates in the length and thickness directions, respectively. In the Timoshenko beam theory, the beam is considered as a structure with three degrees of freedom. Thus, the strain vector in terms of the vector of degrees of freedom can be expressed as

$$\mathbf{U}(x_1, x_3) = \begin{bmatrix} 1 & 0 & x_3 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} u(x_1) \\ w(x_1) \\ \psi(x_1) \end{bmatrix} = \mathbf{P}_0(x_3) \mathbf{q}(x_1), \quad (5.2)$$

in which $\mathbf{P}_0(x_3)$ and $\mathbf{q}(x_1)$ are as follows

$$\mathbf{q}(x_1) = \begin{bmatrix} u(x_1) \\ w(x_1) \\ \psi(x_1) \end{bmatrix}, \quad (5.3)$$

$$\mathbf{P}_0(x_3) = \begin{bmatrix} 1 & 0 & x_3 \\ 0 & 1 & 0 \end{bmatrix}. \quad (5.4)$$

Considering the von Kármán strains, the nonlinear strain vector is constructed as

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \tilde{\varepsilon}_{11} \\ 2\tilde{\varepsilon}_{13} \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x_1} + x_3 \frac{\partial \psi}{\partial x_1} + \frac{1}{2} \left(\frac{\partial w}{\partial x_1} \right)^2 \\ \psi + \frac{\partial w}{\partial x_1} \end{bmatrix}, \quad (5.5)$$

which can be expressed in terms of the vector of degrees of freedom as

$$\boldsymbol{\varepsilon} = \mathbf{P}_1(x_3) \mathbf{E} \mathbf{q}(x_1) + \frac{1}{2} (\mathbf{G} \mathbf{q}(x_1)) \circ (\mathbf{G} \mathbf{q}(x_1)), \quad (5.6)$$

where

$$\mathbf{P}_1(x_3) = \begin{bmatrix} 1 & x_3 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (5.7)$$

$$\mathbf{E} = \begin{bmatrix} \partial/\partial x_1 & 0 & 0 \\ 0 & 0 & \partial/\partial x_1 \\ 0 & \partial/\partial x_1 & 1 \end{bmatrix}, \quad (5.8)$$

$$\mathbf{G} = \begin{bmatrix} 0 & \frac{\partial}{\partial x_1} & 0 \\ 0 & 0 & 0 \end{bmatrix}. \quad (5.9)$$

Since one can write $\mathbf{X} \circ \mathbf{Y} = \langle \mathbf{X} \rangle \mathbf{Y} = \langle \mathbf{Y} \rangle \mathbf{X}$, the strain vector can be given by

$$\boldsymbol{\varepsilon} = \mathbf{P}_1(x_3) \mathbf{E} \mathbf{q}(x_1) + \frac{1}{2} \mathbf{E}_n \mathbf{q}(x_1), \quad (5.10)$$

in which the symmetric matrix $\mathbf{E}_n$ is constructed as

$$\mathbf{E}_n = \langle \mathbf{G} \mathbf{q}(x_1) \rangle \mathbf{G}. \quad (5.11)$$

To write Hamilton's principle for the Timoshenko beam, one must obtain the expressions of kinetic energy, strain energy and work of external forces. To this end, using Eq. (5.2), the kinetic energy and external work are written as

$$T = \frac{1}{2} \int_{\Omega} \dot{\mathbf{U}}^T \rho \dot{\mathbf{U}} dV = \frac{1}{2} \int_{\Omega} \dot{\mathbf{q}}^T(x_1) \mathbf{P}_0^T(x_3) \rho \mathbf{P}_0(x_3) \dot{\mathbf{q}}(x_1) dV, \quad (5.12)$$

$$W_{ext} = \int_{\Omega} \mathbf{U}^T \tilde{\mathbf{F}} dV = \int_{\Omega} \mathbf{q}^T(x_1) \mathbf{P}_0^T(x_3) \tilde{\mathbf{F}} dV, \quad (5.13)$$

where $\tilde{\mathbf{F}}$ denotes the external forces applied on the beam. Since the vector $\mathbf{q}(x_1)$ is not a function of x_2 and x_3 , one can express Eqs. (5.12) and (5.13) as follows

$$T = \frac{1}{2} \int_0^L \dot{\mathbf{q}}^T(x_1) \rho \dot{\mathbf{q}}(x_1) dx_1, \quad (5.14)$$

$$W_{ext} = \int_0^L \mathbf{q}^T(x_1) \tilde{\mathbf{F}} dx_1, \quad (5.15)$$

in which

$$\rho = \int_{-\frac{b}{2}}^{\frac{b}{2}} \int_{-\frac{h}{2}}^{\frac{h}{2}} \mathbf{P}_0^T(x_3) \rho \mathbf{P}_0(x_3) dx_3 dx_2, \quad (5.16)$$

$$\tilde{\mathbf{F}} = \int_{-\frac{b}{2}}^{\frac{b}{2}} \int_{-\frac{h}{2}}^{\frac{h}{2}} \mathbf{P}_0^T(x_3) \tilde{\mathbf{F}} dx_3 dx_2. \quad (5.17)$$

For a homogenous and isotropic beam, density is independent of coordinates. Hence, the matrix ρ can be formulated as

$$\rho = \begin{bmatrix} \rho A & 0 & 0 \\ 0 & \rho A & 0 \\ 0 & 0 & \rho I \end{bmatrix}, \quad (5.18)$$

where $A=bh$ is the area of cross-section of the beam and $I=bh^3/12$ denotes its second moment of area with respect to x_2 axis in x_2x_3 -plane.

Now, one can obtain the strain energy as follows using the strain obtained in Eq. (5.10)

$$\begin{aligned}
 U &= \frac{1}{2} \int_{\Omega} \boldsymbol{\varepsilon}^T \mathbf{C} \boldsymbol{\varepsilon} dV = \frac{1}{2} \int_{\Omega} \mathbf{q}^T(x_1) \left(\mathbf{E}^T \mathbf{P}_1^T(x_3) + \frac{1}{2} \mathbf{E}_n^T \right) \mathbf{C} \left(\mathbf{P}_1(x_3) \mathbf{E} + \frac{1}{2} \mathbf{E}_n \right) \mathbf{q}(x_1) dV \\
 &= \frac{1}{2} \int_{\Omega} \mathbf{q}^T(x_1) \left(\mathbf{E}^T \mathbf{P}_1^T(x_3) \mathbf{C} \mathbf{P}_1(x_3) \mathbf{E} + \frac{1}{2} \mathbf{E}^T \mathbf{P}_1^T(x_3) \mathbf{C} \mathbf{E}_n + \frac{1}{2} \mathbf{E}_n^T \mathbf{C} \mathbf{P}_1(x_3) \mathbf{E} + \frac{1}{4} \mathbf{E}_n^T \mathbf{C} \mathbf{E}_n \right) \mathbf{q}(x_1) dV.
 \end{aligned} \quad (5.19)$$

In this equation, the matrix $\mathbf{C}$ for a homogenous and isotropic beam is given by

$$\mathbf{C} = \begin{bmatrix} \lambda + 2\mu & 0 \\ 0 & \mu \end{bmatrix}. \quad (5.20)$$

As $\mathbf{q}(x_1)$ is not dependent on x_2 and x_3 , the strain energy can be rewritten as

$$U = \frac{1}{2} \int_0^L \mathbf{q}^T(x_1) \left(\mathbf{E}^T \mathbf{C}_1 \mathbf{E} + \frac{1}{2} \mathbf{E}^T \mathbf{C}_2 \mathbf{E}_n + \frac{1}{2} \mathbf{E}_n^T \mathbf{C}_3 \mathbf{E} + \frac{1}{4} \mathbf{E}_n^T \mathbf{C}_4 \mathbf{E}_n \right) \mathbf{q}(x_1) dx_1, \quad (5.21)$$

where matrices $\mathbf{C}_1$ to $\mathbf{C}_4$ are defined as

$$\mathbf{C}_1 = \int_{-\frac{b}{2}}^{\frac{b}{2}} \int_{-\frac{b}{2}}^{\frac{b}{2}} \mathbf{P}_1^T(x_3) \mathbf{C} \mathbf{P}_1(x_3) dx_3 dx_2, \quad (5.22)$$

$$\mathbf{C}_2 = \int_{-\frac{b}{2}}^{\frac{b}{2}} \int_{-\frac{b}{2}}^{\frac{b}{2}} \mathbf{P}_1^T(x_3) \mathbf{C} dx_3 dx_2, \quad (5.23)$$

$$\mathbf{C}_3 = \int_{-\frac{b}{2}}^{\frac{b}{2}} \int_{-\frac{b}{2}}^{\frac{b}{2}} \mathbf{C} \mathbf{P}_1(x_3) dx_3 dx_2 = \mathbf{C}_2^T, \quad (5.24)$$

$$\mathbf{C}_4 = \int_{-\frac{b}{2}}^{\frac{b}{2}} \int_{-\frac{b}{2}}^{\frac{b}{2}} \mathbf{C} dx_3 dx_2. \quad (5.25)$$

For a homogenous and isotropic beam, the matrix $\mathbf{C}$ is independent of coordinates, and hence the following relations are obtained

$$\mathbf{C}_1 = \begin{bmatrix} (\lambda + 2\mu)A & 0 & 0 \\ 0 & (\lambda + 2\mu)I & 0 \\ 0 & 0 & \mu A \end{bmatrix}, \quad (5.26)$$

$$\mathbf{C}_2 = \begin{bmatrix} (\lambda + 2\mu)A & 0 \\ 0 & 0 \\ 0 & \mu A \end{bmatrix} = \mathbf{C}_3^T, \quad (5.27)$$

$$\mathbf{C}_4 = A\mathbf{C}. \quad (5.28)$$

To obtain the discretized equation of motion, one needs to discretize the vector $\mathbf{q}(x_3)$ as follows

$$\mathbf{q} = \begin{bmatrix} \mathbf{u} \\ \mathbf{w} \\ \boldsymbol{\Psi} \end{bmatrix}, \quad (5.29)$$

where $\mathbf{u}$, $\mathbf{w}$ and $\boldsymbol{\Psi}$ are the values of $u(x_1)$, $w(x_1)$ and $\psi(x_1)$ at the grid points. Let $\tilde{\mathbf{S}}_{x_1}$ be the integral operator along the x_1 direction, and

$$\mathbf{S}_{x_1} = \text{diag}(\tilde{\mathbf{S}}_{x_1}) =: \langle \tilde{\mathbf{S}}_{x_1} \rangle. \quad (5.30)$$

Using the Vdq method and in accordance with Eqs. (5.14) and (5.15), the kinetic energy and the external work for Timoshenko beams are derived as

$$T = \frac{1}{2} \dot{\mathbf{q}}^T (\boldsymbol{\rho} \otimes \mathbf{S}_{x_1}) \dot{\mathbf{q}} = \frac{1}{2} \dot{\mathbf{q}}^T \mathbb{M} \dot{\mathbf{q}}, \quad (5.31)$$

$$W_{ext} = \mathbf{q}^T (\mathbf{I}_3 \otimes \mathbf{S}_{x_1}) \bar{\mathbf{F}} = \mathbf{q}^T \mathbb{F}, \quad (5.32)$$

in which the vector $\bar{\mathbf{F}}$ is the discretized counterpart of the vector $\bar{\mathbf{F}}$. Also, matrices $\mathbb{M}$ and $\mathbb{F}$ are defined as

$$\mathbb{M} = \boldsymbol{\rho} \otimes \mathbf{S}_{x_1}, \quad (5.33)$$

$$\mathbb{F} = (\mathbf{I}_3 \otimes \mathbf{S}_{x_1}) \bar{\mathbf{F}}. \quad (5.34)$$

Using the relation $\mathbf{V}_1^T \mathbf{A} \mathbf{V}_2 = \mathbf{V}_2^T \mathbf{A}^T \mathbf{V}_1$, the variations of T and W_{ext} are calculated as follows

$$\delta T = \frac{1}{2} (\delta \dot{\mathbf{q}}^T \mathbb{M} \dot{\mathbf{q}} + \dot{\mathbf{q}}^T \mathbb{M} \delta \dot{\mathbf{q}}) = \delta \dot{\mathbf{q}}^T \mathbb{M} \dot{\mathbf{q}}, \quad (5.35)$$

$$\delta W_{ext} = \delta \mathbf{q}^T \mathbb{F}. \quad (5.36)$$

Similarly, using the Vdq method, one can discretize the total strain energy given by Eq. (5.21), as

$$U = \frac{1}{2} \mathbf{q}^T \left(\mathbb{E}^T (\mathbf{C}_1 \otimes \mathbf{S}_{x_1}) \mathbb{E} + \frac{1}{2} \mathbb{E}^T (\mathbf{C}_2 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n + \frac{1}{2} \mathbb{E}_n^T (\mathbf{C}_2^T \otimes \mathbf{S}_{x_1}) \mathbb{E} + \frac{1}{4} \mathbb{E}_n^T (\mathbf{C}_4 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n \right) \mathbf{q}, \quad (5.37)$$

where matrices $\mathbb{E}$ and $\mathbb{E}_n$ are defined as

$$\mathbb{E} = \begin{bmatrix} \mathbf{D}_{x_1}^{(1)} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{D}_{x_1}^{(1)} \\ \mathbf{0} & \mathbf{D}_{x_1}^{(1)} & \mathbf{I}_n \end{bmatrix}, \quad (5.38)$$

$$\mathbb{E}_n = \langle \mathbb{G} \mathbf{q} \rangle \mathbb{G}, \quad (5.39)$$

$$\mathbb{G} = \begin{bmatrix} \mathbf{0} & \mathbf{D}_{x_1}^{(1)} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}. \quad (5.40)$$

In these relations, $\mathbf{D}_{x_1}^{(1)}$ is the GDQ differential operator.

So as to calculate the variation of Eq. (5.37), one can obtain the variation of every term individually and then add them together. Based on Eq. (5.38), $\mathbb{E}$ is not dependent on $\mathbf{q}$. Thus,

$$\delta(\mathbb{E} \mathbf{q}) = \mathbb{E} \delta \mathbf{q}. \quad (5.41)$$

However, according to Eq. (5.39),

$$\delta(\mathbb{E}_n \mathbf{q}) = \delta \mathbb{E}_n \mathbf{q} + \mathbb{E}_n \delta \mathbf{q} = \langle \mathbb{G} \delta \mathbf{q} \rangle \mathbb{G} \mathbf{q} + \mathbb{E}_n \delta \mathbf{q} = \langle \mathbb{G} \mathbf{q} \rangle \mathbb{G} \delta \mathbf{q} + \mathbb{E}_n \delta \mathbf{q} = 2 \mathbb{E}_n \delta \mathbf{q}. \quad (5.42)$$

Now, using the relation $(\mathbf{L}_1 \otimes \mathbf{L}_2)^T = \mathbf{L}_1^T \otimes \mathbf{L}_2^T$, and bringing to mind that matrices $\mathbf{C}_1$ and $\mathbf{C}_4$ and the diagonal matrix $\mathbf{S}_{x_1}$ are symmetric, one can achieve the following relations

$$\delta(\mathbf{q}^T \mathbb{E}^T (\mathbf{C}_1 \otimes \mathbf{S}_{x_1}) \mathbb{E} \mathbf{q}) = \delta \mathbf{q}^T \mathbb{E}^T (\mathbf{C}_1 \otimes \mathbf{S}_{x_1}) \mathbb{E} \mathbf{q} + \mathbf{q}^T \mathbb{E}^T (\mathbf{C}_1 \otimes \mathbf{S}_{x_1}) \mathbb{E} \delta \mathbf{q} = 2 \delta \mathbf{q}^T \mathbb{E}^T (\mathbf{C}_1 \otimes \mathbf{S}_{x_1}) \mathbb{E} \mathbf{q}, \quad (5.43)$$

$$\begin{aligned} \delta(\mathbf{q}^T \mathbb{E}_n^T (\mathbf{C}_4 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n \mathbf{q}) &= \delta(\mathbb{E}_n \mathbf{q})^T (\mathbf{C}_4 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n \mathbf{q} + \mathbf{q}^T \mathbb{E}_n^T (\mathbf{C}_4 \otimes \mathbf{S}_{x_1}) \delta(\mathbb{E}_n \mathbf{q}) = 2 \delta(\mathbb{E}_n \mathbf{q})^T (\mathbf{C}_4 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n \mathbf{q} \\ &= 4 \delta \mathbf{q}^T \mathbb{E}_n^T (\mathbf{C}_4 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n \mathbf{q}. \end{aligned} \quad (5.44)$$

$$\begin{aligned} \delta(\mathbf{q}^T \mathbb{E}^T (\mathbf{C}_2 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n \mathbf{q}) &= \delta \mathbf{q}^T \mathbb{E}^T (\mathbf{C}_2 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n \mathbf{q} + \mathbf{q}^T \mathbb{E}^T (\mathbf{C}_2 \otimes \mathbf{S}_{x_1}) \delta(\mathbb{E}_n \mathbf{q}) \\ &= \delta \mathbf{q}^T \mathbb{E}^T (\mathbf{C}_2 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n \mathbf{q} + \delta(\mathbb{E}_n \mathbf{q})^T (\mathbf{C}_2^T \otimes \mathbf{S}_{x_1}) \mathbb{E} \mathbf{q} \\ &= \delta \mathbf{q}^T \mathbb{E}^T (\mathbf{C}_2 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n \mathbf{q} + 2 \delta \mathbf{q}^T \mathbb{E}_n^T (\mathbf{C}_2^T \otimes \mathbf{S}_{x_1}) \mathbb{E} \mathbf{q}, \end{aligned} \quad (5.45)$$

$$\begin{aligned} \delta(\mathbf{q}^T \mathbb{E}_n^T (\mathbf{C}_2^T \otimes \mathbf{S}_{x_1}) \mathbb{E} \mathbf{q}) &= \delta(\mathbf{q}^T \mathbb{E}_n^T) (\mathbf{C}_2^T \otimes \mathbf{S}_{x_1}) \mathbb{E} \mathbf{q} + \mathbf{q}^T \mathbb{E}_n^T (\mathbf{C}_2^T \otimes \mathbf{S}_{x_1}) \delta \mathbb{E} \mathbf{q} \\ &= 2 \delta \mathbf{q}^T \mathbb{E}_n^T (\mathbf{C}_2^T \otimes \mathbf{S}_{x_1}) \mathbb{E} \mathbf{q} + \delta \mathbf{q}^T \mathbb{E}^T (\mathbf{C}_2 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n \mathbf{q}. \end{aligned} \quad (5.46)$$

Using the above relations, the variation of the discretized form of strain energy given by Eq. (5.37) becomes

$$\delta U = \delta \mathbf{q}^T \left(\mathbb{E}^T (\mathbf{C}_1 \otimes \mathbf{S}_{x_1}) \mathbb{E} + \frac{1}{2} (\mathbb{E}^T (\mathbf{C}_2 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n + 2 \mathbb{E}_n^T (\mathbf{C}_2^T \otimes \mathbf{S}_{x_1}) \mathbb{E}) + \frac{1}{2} \mathbb{E}_n^T (\mathbf{C}_4 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n \right) \mathbf{q}. \quad (5.47)$$

It is also possible to directly discretize the variation of strain energy instead of calculating the variation of the discretized form of strain energy. Since the variation of strain energy is given by

$$\begin{aligned} \delta U &= \int_{\Omega} \delta \boldsymbol{\varepsilon}^T \mathbf{C} \boldsymbol{\varepsilon} dV = \int_{\Omega} \delta \mathbf{q}^T(x_1) (\mathbf{E}^T \mathbf{P}_1^T(x_3) + \mathbf{E}_n^T) \mathbf{C} (\mathbf{P}_1(x_3) \mathbf{E} + \frac{1}{2} \mathbf{E}_n) \mathbf{q}(x_1) dV \\ &= \int_0^L \delta \mathbf{q}^T(x_1) \left(\mathbf{E}^T \mathbf{C}_1 \mathbf{E} + \frac{1}{2} (\mathbf{E}^T \mathbf{C}_2 \mathbf{E}_n + 2 \mathbf{E}_n^T \mathbf{C}_2^T \mathbf{E}) + \frac{1}{2} \mathbf{E}_n^T \mathbf{C}_4 \mathbf{E}_n \right) \mathbf{q}(x_1) dx_1. \end{aligned} \quad (5.48)$$

Its discretized form exactly becomes the same as Eq. (5.47). One can write Eq. (5.47) as

$$\delta U = \delta \mathbf{q}^T \mathbb{K}_s \mathbf{q}. \quad (5.49)$$

In this relation, $\mathbb{K}_s$ is the symmetrized secant stiffness matrix (see Eqs. (4.67)–(4.77)), which can be defined as

$$\mathbb{K}_s = \mathbb{K} + \frac{1}{2} \mathbb{N}_1 + \frac{1}{3} \mathbb{N}_2, \quad (5.50)$$

where

$$\mathbb{K} = \mathbb{E}^T (\mathbf{C}_1 \otimes \mathbf{S}_{x_1}) \mathbb{E}, \quad (5.51)$$

$$\mathbb{N}_1 = \mathbb{E}^T (\mathbf{C}_2 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n + \mathbb{E}_n^T (\mathbf{C}_2^T \otimes \mathbf{S}_{x_1}) \mathbb{E} + \mathbb{K}_1, \quad (5.52)$$

$$\mathbb{K}_1 = \mathbb{G}^T \langle (\mathbf{C}_2^T \otimes \mathbf{S}_{x_1}) \mathbb{E} \mathbf{q} \rangle \mathbb{G}, \quad (5.53)$$

$$\mathbb{N}_2 = \frac{3}{2} \mathbb{E}_n^T (\mathbf{C}_4 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n. \quad (5.54)$$

Consequently, using Eqs. (5.35), (5.36) and (5.49), the discretized representation of Hamilton's principle becomes

$$\int_{t_1}^{t_2} (\delta T - \delta U + \delta W_{ext}) dt = \int_{t_1}^{t_2} (\delta \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}} - \delta \mathbf{q}^T \mathbf{K}_s \mathbf{q} + \delta \mathbf{q}^T \mathbf{F}) dt = 0. \quad (5.55)$$

Integrating by parts results in

$$\int_{t_1}^{t_2} \delta T dt = \delta \mathbf{q}^T \mathbf{M} \dot{\mathbf{q}} \Big|_{t_1}^{t_2} - \int_{t_1}^{t_2} \delta \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}} dt = \delta \mathbf{q}^T(t_2) \mathbf{M} \dot{\mathbf{q}}(t_2) - \delta \mathbf{q}^T(t_1) \mathbf{M} \dot{\mathbf{q}}(t_1) - \int_{t_1}^{t_2} \delta \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}} dt. \quad (5.56)$$

Since Hamilton's principle is written for a time interval between two moments with definite status, the variation of displacement vector at those statuses is equal to zero. Thus, $\delta \mathbf{q}(t_2) = \delta \mathbf{q}(t_1) = 0$ which leads to

$$\int_{t_1}^{t_2} \delta T dt = - \int_{t_1}^{t_2} \delta \dot{\mathbf{q}}^T \mathbf{M} \dot{\mathbf{q}} dt. \quad (5.57)$$

Hence

$$\int_{t_1}^{t_2} (\delta T - \delta U + \delta W_{ext}) dt = \int_{t_1}^{t_2} \delta \mathbf{q}^T (-\mathbf{M} \ddot{\mathbf{q}} - \mathbf{K}_s \mathbf{q} + \mathbf{F}) dt = 0. \quad (5.58)$$

Using the fundamental lemma of calculus of variation, the discretized equation of motion is derived as

$$\mathbf{M} \ddot{\mathbf{q}} + \mathbf{K}_s \mathbf{q} - \mathbf{F} = \mathbf{0}. \quad (5.59)$$

Solution of this equation by means of the Newton–Raphson scheme leads to finding the displacement values at grid points. Also, for a more general case, we demonstrated in Section 4.4 that the tangent stiffness matrix $\mathbf{K}_t = \mathbf{K} + \mathbf{N}_1 + \mathbf{N}_2$.

Let $0 \leq x_1 \leq L$ where L is the length of the beam, then the following cases of boundary conditions can be considered for the beam:

- Clamped (CC): $u=w=\psi=0$ at $x_1=0, L$,
- Simply-Supported (SS): $u=w=0$ at $x_1=0, L$,
- Cantilever (CF): $u=w=\psi=0$ at $x_1=0$.

6. VDQ formulation for the Mindlin plate theory

Based on Mindlin plate theory, also known as the first-order shear deformation plate theory (FSDT), the displacement field of plate is given by

$$\begin{cases} u_1(x_1, x_2, x_3) = u(x_1, x_2) + x_3 \psi(x_1, x_2), \\ u_2(x_1, x_2, x_3) = v(x_1, x_2) + x_3 \phi(x_1, x_2), \\ u_3(x_1, x_2, x_3) = w(x_1, x_2), \end{cases} \quad (6.1)$$

which can be shown as

$$\mathbf{U}(x_1, x_2, x_3) = \begin{bmatrix} 1 & 0 & 0 & x_3 & 0 \\ 0 & 1 & 0 & 0 & x_3 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} u(x_1, x_2) \\ v(x_1, x_2) \\ w(x_1, x_2) \\ \psi(x_1, x_2) \\ \phi(x_1, x_2) \end{bmatrix} = \mathbf{P}_0(x_3) \mathbf{q}(x_1, x_2), \quad (6.2)$$

where

$$\mathbf{q}(x_1, x_2) = \begin{bmatrix} u(x_1, x_2) \\ v(x_1, x_2) \\ w(x_1, x_2) \\ \psi(x_1, x_2) \\ \phi(x_1, x_2) \end{bmatrix}, \quad (6.3)$$

$$\mathbf{P}_0(x_3) = \begin{bmatrix} 1 & 0 & 0 & x_3 & 0 \\ 0 & 1 & 0 & 0 & x_3 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix}. \quad (6.4)$$

Using the von Kármán strains, the following relation for the nonlinear strains of Mindlin plate can be written

$$\boldsymbol{\varepsilon} = \begin{bmatrix} \tilde{\varepsilon}_{11}(x_1, x_2, x_3) \\ \tilde{\varepsilon}_{22}(x_1, x_2, x_3) \\ 2\tilde{\varepsilon}_{12}(x_1, x_2, x_3) \\ 2\tilde{\varepsilon}_{13}(x_1, x_2) \\ 2\tilde{\varepsilon}_{23}(x_1, x_2) \end{bmatrix} = \begin{bmatrix} \frac{\partial u}{\partial x_1} + x_3 \frac{\partial \psi}{\partial x_1} + \frac{1}{2} \left(\frac{\partial w}{\partial x_1} \right)^2 \\ \frac{\partial v}{\partial x_2} + x_3 \frac{\partial \phi}{\partial x_2} + \frac{1}{2} \left(\frac{\partial w}{\partial x_2} \right)^2 \\ \frac{\partial u}{\partial x_2} + \frac{\partial v}{\partial x_1} + x_3 \left(\frac{\partial \psi}{\partial x_2} + \frac{\partial \phi}{\partial x_1} \right) + \frac{\partial w}{\partial x_1} \frac{\partial w}{\partial x_2} \\ \psi + \frac{\partial w}{\partial x_1} \\ \phi + \frac{\partial w}{\partial x_2} \end{bmatrix}, \quad (6.5)$$

which can be rewritten as

$$\mathcal{E} = \mathbf{P}_1(x_3)\mathbf{E}\mathbf{q}(x_1, x_2) + (\mathbf{G}_1\mathbf{q}(x_1, x_2)) \circ (\mathbf{G}_2\mathbf{q}(x_1, x_2)), \quad (6.6)$$

where

$$\mathbf{P}_1(x_3) = \begin{bmatrix} 1 & 0 & 0 & x_3 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & x_3 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & x_3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}, \quad (6.7)$$

$$\mathbf{E} = \begin{bmatrix} \partial/\partial x_1 & 0 & 0 & 0 & 0 & 0 \\ 0 & \partial/\partial x_2 & 0 & 0 & 0 & 0 \\ \partial/\partial x_2 & \partial/\partial x_1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & \partial/\partial x_1 & 0 & 0 \\ 0 & 0 & 0 & 0 & \partial/\partial x_2 & 0 \\ 0 & 0 & 0 & \partial/\partial x_2 & \partial/\partial x_1 & 0 \\ 0 & 0 & \partial/\partial x_1 & 1 & 0 & 0 \\ 0 & 0 & \partial/\partial x_2 & 0 & 1 & 0 \end{bmatrix}, \quad (6.8)$$

$$\mathbf{G}_1 = \begin{bmatrix} 0 & 0 & \partial/\partial x_1 & 0 & 0 \\ 0 & 0 & \partial/\partial x_2 & 0 & 0 \\ 0 & 0 & \partial/\partial x_1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}, \quad (6.9)$$

$$\mathbf{G}_2 = \begin{bmatrix} 0 & 0 & \frac{1}{2} \frac{\partial}{\partial x_1} & 0 & 0 \\ 0 & 0 & \frac{1}{2} \frac{\partial}{\partial x_2} & 0 & 0 \\ 0 & 0 & \frac{\partial}{\partial x_2} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}. \quad (6.10)$$

Now, using the relation $\mathbf{X} \circ \mathbf{Y} = \langle \mathbf{X} \rangle \mathbf{Y} = \langle \mathbf{Y} \rangle \mathbf{X} = \frac{1}{2} (\langle \mathbf{X} \rangle \mathbf{Y} + \langle \mathbf{Y} \rangle \mathbf{X})$, the strain vector is

$$\mathcal{E} = \mathbf{P}_1(x_3)\mathbf{E}\mathbf{q}(x_1, x_2) + \frac{1}{2}\mathbf{E}_n\mathbf{q}(x_1, x_2), \quad (6.11)$$

where

$$\mathbf{E}_n = \langle \mathbf{G}_1\mathbf{q}(x_1, x_2) \rangle \mathbf{G}_2 + \langle \mathbf{G}_2\mathbf{q}(x_1, x_2) \rangle \mathbf{G}_1. \quad (6.12)$$

For Mindlin plate, the kinetic energy and the external work are formulated as

$$T = \frac{1}{2} \int_{\Omega} \dot{\mathbf{U}}^T \rho \dot{\mathbf{U}} dV = \frac{1}{2} \int_{\Omega} \dot{\mathbf{q}}^T(x_1, x_2) \mathbf{P}_0^T(x_3) \rho \mathbf{P}_0(x_3) \dot{\mathbf{q}}(x_3, x_2) dV, \quad (6.13)$$

$$W_{ext} = \int_{\Omega} \mathbf{U}^T \tilde{\mathbf{F}} dV = \int_{\Omega} \mathbf{q}^T(x_1, x_2) \mathbf{P}_0^T(x_3) \tilde{\mathbf{F}} dV, \quad (6.14)$$

where $\tilde{\mathbf{F}}$ denotes the external forces applied on the plate. Since the vector $\mathbf{q}(x_1, x_2)$ is not a function of x_3 , Eqs. (6.13) and (6.14) can be written as

$$T = \frac{1}{2} \int_0^a \int_0^b \dot{\mathbf{q}}^T(x_1, x_2) \rho \dot{\mathbf{q}}(x_1, x_2) dx_2 dx_1, \quad (6.15)$$

$$W_{ext} = \int_0^a \int_0^b \mathbf{q}^T(x_1, x_2) \tilde{\mathbf{F}} dx_2 dx_1, \quad (6.16)$$

where

$$\rho = \int_{-\frac{h}{2}}^{\frac{h}{2}} \mathbf{P}_0^T(x_3) \rho \mathbf{P}_0(x_3) dx_3, \quad (6.17)$$

$$\tilde{\mathbf{F}} = \int_{-\frac{h}{2}}^{\frac{h}{2}} \mathbf{P}_0^T(x_3) \tilde{\mathbf{F}} dx_3. \quad (6.18)$$

For a homogeneous and isotropic plate, the density is not dependent on the coordinates. Accordingly, by indicating the thickness of plate as h , the matrix ρ is obtained as

$$\rho = \rho \begin{bmatrix} h\mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & \frac{h^3}{12}\mathbf{I}_2 \end{bmatrix}, \quad (6.19)$$

where $\mathbf{I}_k$ is a $k \times k$ identity matrix.

Using the strain obtained from Eq. (6.11), the strain energy is written as

$$\begin{aligned} U &= \frac{1}{2} \int_{\Omega} \boldsymbol{\varepsilon}^T \mathbf{C} \boldsymbol{\varepsilon} dV \\ &= \frac{1}{2} \int_{\Omega} \mathbf{q}^T(x_1, x_2) \left(\mathbf{E}^T \mathbf{P}_1^T(x_3) + \frac{1}{2} \mathbf{E}_n^T \right) \mathbf{C} \left(\mathbf{P}_1(x_3) \mathbf{E} + \frac{1}{2} \mathbf{E}_n \right) \mathbf{q}(x_1, x_2) dV \\ &= \frac{1}{2} \int_{\Omega} \mathbf{q}^T(x_1, x_2) \left(\mathbf{E}^T \mathbf{P}_1^T(x_3) \mathbf{C} \mathbf{P}_1(x_3) \mathbf{E} + \frac{1}{2} \mathbf{E}^T \mathbf{P}_1^T(x_3) \mathbf{C} \mathbf{E}_n + \frac{1}{2} \mathbf{E}_n^T \mathbf{C} \mathbf{P}_1(x_3) \mathbf{E} + \frac{1}{4} \mathbf{E}_n^T \mathbf{C} \mathbf{E}_n \right) \mathbf{q}(x_1, x_2) dV. \end{aligned} \quad (6.20)$$

For a homogenous and isotropic plate, the matrix $\mathbf{C}$ is obtained as

$$\mathbf{C} = \begin{bmatrix} \mathbf{C}_{in} & \mathbf{0} \\ \mathbf{0} & \mathbf{C}_{out} \end{bmatrix}, \quad (6.21)$$

where $\mathbf{C}_{in}$ and $\mathbf{C}_{out}$ are the elasticity stiffness matrices of in-plane and out-of-plane strains, respectively, which are formulated as

$$\mathbf{C}_{in} = \begin{bmatrix} \lambda + 2\mu & \lambda & 0 \\ \lambda & \lambda + 2\mu & 0 \\ 0 & 0 & \mu \end{bmatrix}, \quad (6.22)$$

$$\mathbf{C}_{out} = \begin{bmatrix} k_s \mu & 0 \\ 0 & k_s \mu \end{bmatrix}, \quad (6.23)$$

where k_s is the shear correction factor. Accordingly, Eq. (6.20) can be written as

$$U = \frac{1}{2} \int_0^a \int_0^b \mathbf{q}^T(x_1, x_2) \left(\mathbf{E}^T \mathbf{C}_1 \mathbf{E} + \frac{1}{2} \mathbf{E}^T \mathbf{C}_2 \mathbf{E}_n + \frac{1}{2} \mathbf{E}_n^T \mathbf{C}_3 \mathbf{E} + \frac{1}{4} \mathbf{E}_n^T \mathbf{C}_4 \mathbf{E}_n \right) \mathbf{q}(x_1, x_2) dx_2 dx_1, \quad (6.24)$$

where

$$\mathbf{C}_1 = \int_{-\frac{h}{2}}^{\frac{h}{2}} \mathbf{P}_1^T(x_3) \mathbf{C} \mathbf{P}_1(x_3) dx_3, \quad (6.25)$$

$$\mathbf{C}_2 = \int_{-\frac{h}{2}}^{\frac{h}{2}} \mathbf{P}_1^T(x_3) \mathbf{C} dx_3, \quad (6.26)$$

$$\mathbf{C}_3 = \int_{-\frac{h}{2}}^{\frac{h}{2}} \mathbf{C} \mathbf{P}_1(x_3) dx_3 = \mathbf{C}_2^T, \quad (6.27)$$

$$\mathbf{C}_4 = \int_{-\frac{h}{2}}^{\frac{h}{2}} \mathbf{C} dx_3. \quad (6.28)$$

For a homogenous and isotropic plate, these matrices are calculated by

$$\mathbf{C}_1 = \begin{bmatrix} \mathcal{A} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathcal{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & h \mathbf{C}_{out} \end{bmatrix}, \quad (6.29)$$

$$\mathbf{C}_2 = \begin{bmatrix} \mathcal{A} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \\ \mathbf{0} & h \mathbf{C}_{out} \end{bmatrix}_{8 \times 5} = \mathbf{C}_3^T, \quad (6.30)$$

$$\mathbf{C}_4 = h \mathbf{C}. \quad (6.31)$$

The matrix $\mathbf{C}_{out}$ is given by Eq. (6.23), and based on Eq. (6.22), matrices $\mathcal{A}$ and $\mathcal{D}$ are

$$\mathcal{A} = h \mathbf{C}_{in}, \quad (6.32)$$

$$\mathcal{D} = \frac{h^3}{12} \mathbf{C}_{in}. \quad (6.33)$$

It should be noted that for a composite plate, the matrix $\mathbf{C}$ is a function of coordinates, and matrices $\mathbf{C}_1$ to $\mathbf{C}_4$ should be calculated from Eqs. (6.25)–(6.28). In this case, the number of zeros in these matrices decreases. For example, the matrix $\mathbf{C}_1$

becomes $\begin{bmatrix} \mathcal{A} & \mathcal{B} & \mathbf{0} \\ \mathcal{B}^T & \mathcal{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & h \mathbf{C}_{out} \end{bmatrix}$.

By calculating the components of the vector $\mathbf{q}(x_1, x_2)$ at the grid points, one can write each of them as a vector. For example, the nodal values of $u(x_1, x_2)$ can be shown as a column vector $\mathbf{u}$. Hence, using these vectors, $\mathbf{q}$ can be defined as

$$\mathbf{q} = [\mathbf{u}^T \quad \mathbf{v}^T \quad \mathbf{w}^T \quad \boldsymbol{\psi}^T \quad \boldsymbol{\phi}^T]^T. \quad (6.34)$$

Let $\tilde{\mathbb{S}} = \tilde{\mathbb{S}}$ be the integral operator corresponding to the x_1x_2 -plane and let $\mathbb{S} = \langle \tilde{\mathbb{S}} \rangle$. Using the VDQ method. The kinetic energy and the external work are discretized as

$$T = \frac{1}{2} \dot{\mathbf{q}}^T (\boldsymbol{\rho} \otimes \mathbb{S}) \dot{\mathbf{q}} = \frac{1}{2} \dot{\mathbf{q}}^T \mathbb{M} \dot{\mathbf{q}}, \quad (6.35)$$

$$W_{ext} = \mathbf{q}^T (\mathbf{I}_5 \otimes \mathbb{S}) \bar{\mathbf{F}} = \mathbf{q}^T \mathbb{F}, \quad (6.36)$$

where matrices $\mathbb{M}$ and $\mathbb{F}$ are given by

$$\mathbb{M} = \boldsymbol{\rho} \otimes \mathbb{S}, \quad (6.37)$$

$$\mathbb{F} = (\mathbf{I}_5 \otimes \mathbb{S}) \bar{\mathbf{F}}. \quad (6.38)$$

The variations of the kinetic energy and the external work are

$$\delta T = \delta \dot{\mathbf{q}}^T \mathbb{M} \dot{\mathbf{q}}, \quad (6.39)$$

$$\delta W_{ext} = \delta \mathbf{q}^T \mathbb{F}. \quad (6.40)$$

Using the VDQ method, the discretized form of Eq. (6.24) is written as

$$U = \frac{1}{2} \mathbf{q}^T \left(\mathbb{E}^T (\mathbf{C}_1 \otimes \mathbb{S}) \mathbb{E} + \frac{1}{2} \mathbb{E}^T (\mathbf{C}_2 \otimes \mathbb{S}) \mathbb{E}_n + \frac{1}{2} \mathbb{E}_n^T (\mathbf{C}_2^T \otimes \mathbb{S}) \mathbb{E} + \frac{1}{4} \mathbb{E}_n^T (\mathbf{C}_4 \otimes \mathbb{S}) \mathbb{E}_n \right) \mathbf{q}, \quad (6.41)$$

where

$$\mathbb{E} = \begin{bmatrix} \mathbf{I}_{n_2} \otimes \mathbf{D}_{x_1}^{(1)} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{D}_{x_2}^{(1)} \otimes \mathbf{I}_{n_1} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{D}_{x_2}^{(1)} \otimes \mathbf{I}_{n_1} & \mathbf{I}_{n_2} \otimes \mathbf{D}_{x_1}^{(1)} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{I}_{n_2} \otimes \mathbf{D}_{x_1}^{(1)} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{D}_{x_2}^{(1)} \otimes \mathbf{I}_{n_1} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{D}_{x_2}^{(1)} \otimes \mathbf{I}_{n_1} & \mathbf{I}_{n_2} \otimes \mathbf{D}_{x_1}^{(1)} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_{n_2} \otimes \mathbf{D}_{x_1}^{(1)} & \mathbf{I}_{n_2} \otimes \mathbf{I}_{n_1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{D}_{x_2}^{(1)} \otimes \mathbf{I}_{n_1} & \mathbf{0} & \mathbf{I}_{n_2} \otimes \mathbf{I}_{n_1} \end{bmatrix}, \quad (6.42)$$

$$\mathbb{E}_n = \langle \mathbb{G}_1 \mathbf{q} \rangle \mathbb{G}_2 + \langle \mathbb{G}_2 \mathbf{q} \rangle \mathbb{G}_1, \quad (6.43)$$

$$\mathbb{G}_1 = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \mathbf{I}_{n_2} \otimes \mathbf{D}_{x_1}^{(1)} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{D}_{x_2}^{(1)} \otimes \mathbf{I}_{n_1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{I}_{n_2} \otimes \mathbf{D}_{x_1}^{(1)} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}, \quad (6.44)$$

$$\mathbb{G}_2 = \begin{bmatrix} \mathbf{0} & \mathbf{0} & \frac{1}{2} \mathbf{I}_{n_2} \otimes \mathbf{D}_{x_1}^{(1)} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \frac{1}{2} \mathbf{D}_{x_2}^{(1)} \otimes \mathbf{I}_{n_1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{D}_{x_2}^{(1)} \otimes \mathbf{I}_{n_1} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix}. \quad (6.45)$$

In these relations, n_i stands for the number of grid points along the x_i direction, and $\mathbf{D}_{x_i}^{(1)}$ is the GDQ differential operator. Taking variation of $\mathbb{E} \mathbf{q}$ and $\mathbb{E}_n \mathbf{q}$, gives

$$\delta(\mathbb{E} \mathbf{q}) = \mathbb{E} \delta \mathbf{q}, \quad (6.46)$$

$$\delta(\mathbb{E}_n \mathbf{q}) = \delta \mathbb{E}_n \mathbf{q} + \mathbb{E}_n \delta \mathbf{q} = \langle \mathbb{G}_1 \delta \mathbf{q} \rangle \mathbb{G}_2 \mathbf{q} + \langle \mathbb{G}_2 \delta \mathbf{q} \rangle \mathbb{G}_1 \mathbf{q} + \mathbb{E}_n \delta \mathbf{q} = \langle \mathbb{G}_2 \mathbf{q} \rangle \mathbb{G}_1 \delta \mathbf{q} + \langle \mathbb{G}_1 \mathbf{q} \rangle \mathbb{G}_2 \delta \mathbf{q} + \mathbb{E}_n \delta \mathbf{q} = 2 \mathbb{E}_n \delta \mathbf{q}. \quad (6.47)$$

Thus, considering $(\mathbf{L}_1 \otimes \mathbf{L}_2)^T = \mathbf{L}_1^T \otimes \mathbf{L}_2^T$, the variation of strain energy given by Eq. (6.41) is obtained as

$$\delta U = \delta \mathbf{q}^T \left(\mathbb{E}^T (\mathbf{C}_1 \otimes \mathbb{S}) \mathbb{E} + \frac{1}{2} \left(\mathbb{E}^T (\mathbf{C}_2 \otimes \mathbb{S}) \mathbb{E}_n + 2 \mathbb{E}_n^T (\mathbf{C}_2^T \otimes \mathbb{S}) \mathbb{E} \right) + \frac{1}{2} \mathbb{E}_n^T (\mathbf{C}_4 \otimes \mathbb{S}) \mathbb{E}_n \right) \mathbf{q}. \quad (6.48)$$

Note that $\mathbf{C}_1$, $\mathbf{C}_4$, and $\mathbb{S}$ are symmetric, and see Eqs. (5.43)–(5.46). The above equation is written as

$$\delta U = \delta \mathbf{q}^T \mathbb{K}_s \mathbf{q}, \quad (6.49)$$

where the secant stiffness matrix $\mathbb{K}_s$ (see Eqs. (4.67)–(4.77)) is defined as

$$\mathbb{K}_s = \mathbb{K} + \frac{1}{2}\mathbb{N}_1 + \frac{1}{3}\mathbb{N}_2, \quad (6.50)$$

where

$$\mathbb{K} = \mathbb{E}^T(\mathbf{C}_1 \otimes \mathbf{S})\mathbb{E}, \quad (6.51)$$

$$\mathbb{N}_1 = \mathbb{E}^T(\mathbf{C}_2 \otimes \mathbf{S})\mathbb{E}_n + \mathbb{E}_n^T(\mathbf{C}_2^T \otimes \mathbf{S})\mathbb{E} + \mathbb{K}_1, \quad (6.52)$$

$$\mathbb{K}_1 = \mathbf{G}_1^T((\mathbf{C}_2^T \otimes \mathbf{S})\mathbb{E}\mathbf{q})\mathbf{G}_2 + \mathbf{G}_2^T((\mathbf{C}_2^T \otimes \mathbf{S})\mathbb{E}\mathbf{q})\mathbf{G}_1, \quad (6.53)$$

$$\mathbb{N}_2 = \frac{3}{2}\mathbb{E}_n^T(\mathbf{C}_4 \otimes \mathbf{S})\mathbb{E}_n. \quad (6.54)$$

Using Eqs. (6.39), (6.40), and (6.49), the discretized form of Hamilton's principle is expressed as

$$\int_{t_1}^{t_2} (\delta T - \delta U + \delta W_{ext}) dt = \int_{t_1}^{t_2} (\delta \dot{\mathbf{q}}^T \mathbb{M} \dot{\mathbf{q}} - \delta \mathbf{q}^T \mathbb{K}_s \mathbf{q} + \delta \mathbf{q}^T \mathbb{F}) dt = 0. \quad (6.55)$$

Integrating by parts leads to

$$\int_{t_1}^{t_2} \delta T dt = - \int_{t_1}^{t_2} \delta \mathbf{q}^T \mathbb{M} \ddot{\mathbf{q}} dt. \quad (6.56)$$

Hamilton's principle is thus written as

$$\int_{t_1}^{t_2} \delta \mathbf{q}^T (\mathbb{M} \ddot{\mathbf{q}} + \mathbb{K}_s \mathbf{q} - \mathbb{F}) dt = 0 \quad (6.57)$$

Using the fundamental lemma of calculus of variation, the discretized equation of motion is

$$\mathbb{M} \ddot{\mathbf{q}} + \mathbb{K}_s \mathbf{q} - \mathbb{F} = \mathbf{0}. \quad (6.58)$$

Assume $0 \leq x_1 \leq a$ and $0 \leq x_2 \leq b$, where a and b are the length of each edge of the plate. We can consider the following cases of boundary conditions:

- CCCC: $u=v=w=\psi=\varphi=0$ at $x_1=0, a$ and $x_2=0, b$,
- SSSS: $u=v=w=\varphi=0$ at $x_1=0, a$ and $u=v=w=\psi=0$ at $x_2=0, b$,
- CSCS: $u=v=w=\psi=\varphi=0$ at $x_1=0, a$ and $u=v=w=\psi=0$ at $x_2=0, b$,
- CFCF: $u=v=w=\psi=\varphi=0$ at $x_1=0, a$.

C, S, and F denote the simply-supported, clamped, and free edges, respectively.

7. Summary of final equations

Using the VDQ Method, regardless of the type of the problem, the discretized equation of motion is obtained as

$$\mathbb{M} \ddot{\mathbf{U}} + \mathbb{K}_s \mathbf{U} - \mathbb{F} = \mathbf{0},$$

where $\mathbf{U}$ is the vector of unknown degrees of freedom, $\mathbb{M}$ is the mass matrix, $\mathbb{F}$ contains the vectors of external force and body force, and $\mathbb{K}_s$ is the stiffness matrix of the system and is given by

$$\mathbb{K}_s = \mathbb{K} + \frac{1}{2}\mathbb{N}_1(\mathbf{U}) + \frac{1}{3}\mathbb{N}_2(\mathbf{U}).$$

In the above relation, $\mathbb{K}$ is the linear stiffness matrix and is defined based on the linear parts of strain tensor. Also, $\mathbb{N}_1$ and $\mathbb{N}_2$ are the nonlinear parts of the stiffness matrix and correspond to the nonlinear parts of the strain tensor. Due to the formulation of the VDQ method, the matrix $\mathbb{K}_s$ is symmetric.

For the general case of nonlinear elasticity, the mass and the force matrices are obtained as

$$\begin{aligned} \mathbb{M} &= \rho \mathbf{I}_3 \otimes \mathbf{S}, \\ \mathbb{F} &= (\mathbf{I}_3 \otimes \mathbf{S}) \tilde{\mathbb{F}}, \end{aligned}$$

where $\otimes$ denotes the Kronecker product of matrices given by Eq. (4.19), and the diagonal matrix $\mathbf{S}$ is defined using Eq. (4.15) and the integral operators in the Appendix (Section b and Section c). Also, ρ is the density of the material and $\tilde{\mathbb{F}}$ is calculated by Eq. (4.7). Considering the Green-St. Venant strain tensor, the stiffness matrices are

$$\begin{aligned} \mathbb{K} &= \mathbb{E}^T(\mathbf{C} \otimes \mathbf{S})\mathbb{E}, \\ \mathbb{N}_1 &= \mathbb{E}^T(\mathbf{C} \otimes \mathbf{S})\mathbb{E}_n + \mathbb{E}_n^T(\mathbf{C} \otimes \mathbf{S})\mathbb{E} + \sum_{i=1}^3 (\mathbf{G}_{2i-1}^T((\mathbf{C} \otimes \mathbf{S})\mathbb{E}\mathbf{U})\mathbf{G}_{2i} + \mathbf{G}_{2i}^T((\mathbf{C} \otimes \mathbf{S})\mathbb{E}\mathbf{U})\mathbf{G}_{2i-1}), \end{aligned}$$

$$\mathbb{N}_2 = \frac{3}{2} \mathbb{E}_n^T (\mathbf{C} \otimes \mathbb{S}) \mathbb{E}_n,$$

where the symbols $\langle \cdot \rangle$ converts a vector to a diagonal matrix, that is $\langle \cdot \rangle = \text{diag}(\cdot)$ (see Eq. (4.15)). Note that $\mathbf{C}$ denotes the elasticity matrix, which is given by Eq. (3.27). In addition, the matrices $\mathbb{E}$, $\mathbb{E}_n$, and $\mathbb{G}_j$, ($j = 1, 2, \dots, 6$), which are defined based on the Green-St. Venant strain tensor, are obtained by Eqs. (4.40)–(4.46). For calculating these matrices, one needs to use the GDQ differential operators, which are given in the Appendix (Section a and Section c).

Now we summarize the reduced versions of the VDQ method for beams and plates.

For the case of the Timoshenko beam theory, the mass matrix and the force vector are

$$\begin{aligned} \mathbb{M} &= \rho \otimes \mathbf{S}_{x_1}, \\ \mathbb{F} &= (\mathbf{I}_3 \otimes \mathbf{S}_{x_1}) \tilde{\mathbb{F}}, \end{aligned}$$

where ρ is given by Eq. (5.18) and $\tilde{\mathbb{F}}$ is calculated by Eq. (5.17). The one-dimensional integral operator $\mathbf{S}_{x_1}$ is obtained using Eq. (5.30) and Section b of the Appendix. The stiffness matrices of the Timoshenko beam are

$$\begin{aligned} \mathbb{K} &= \mathbb{E}^T (\mathbf{C}_1 \otimes \mathbf{S}_{x_1}) \mathbb{E}, \\ \mathbb{N}_1 &= \mathbb{E}^T (\mathbf{C}_2 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n + \mathbb{E}_n^T (\mathbf{C}_2^T \otimes \mathbf{S}_{x_1}) \mathbb{E} + \mathbb{G}^T \langle (\mathbf{C}_2^T \otimes \mathbf{S}_{x_1}) \mathbb{E}_q \rangle \mathbb{G}, \\ \mathbb{N}_2 &= \frac{3}{2} \mathbb{E}_n^T (\mathbf{C}_4 \otimes \mathbf{S}_{x_1}) \mathbb{E}_n, \end{aligned}$$

where $\mathbf{C}_1$, $\mathbf{C}_2$, and $\mathbf{C}_4$ are given by Eqs. (5.26)–(5.28). The matrices $\mathbb{E}$, $\mathbb{E}_n$, and $\mathbb{G}$ are defined based on the displacement field of the Timoshenko beam theory and the corresponding von Kármán strains. These matrices are defined by Eqs. (5.38)–(5.40) using the GDQ differential operators, which are presented in Section a of the Appendix.

For the case of the Mindlin plate theory, the mass matrix and the force vector are

$$\begin{aligned} \mathbb{M} &= \rho \otimes \mathbb{S}, \\ \mathbb{F} &= (\mathbf{I}_5 \otimes \mathbb{S}) \tilde{\mathbb{F}}, \end{aligned}$$

where ρ is obtained by Eq. (6.19) and $\tilde{\mathbb{F}}$ is calculated using Eq. (6.18). The integral operator is $\mathbb{S} = \langle \tilde{\mathbb{S}} \rangle$, where $\tilde{\mathbb{S}}$ is defined in the Appendix (Section b and Section c). The stiffness matrices of the Mindlin plate are

$$\begin{aligned} \mathbb{K} &= \mathbb{E}^T (\mathbf{C}_1 \otimes \mathbb{S}) \mathbb{E}, \\ \mathbb{N}_1 &= \mathbb{E}^T (\mathbf{C}_2 \otimes \mathbb{S}) \mathbb{E}_n + \mathbb{E}_n^T (\mathbf{C}_2^T \otimes \mathbb{S}) \mathbb{E} + \mathbb{G}_1^T \langle (\mathbf{C}_2^T \otimes \mathbb{S}) \mathbb{E}_q \rangle \mathbb{G}_2 + \mathbb{G}_2^T \langle (\mathbf{C}_2^T \otimes \mathbb{S}) \mathbb{E}_q \rangle \mathbb{G}_1, \\ \mathbb{N}_2 &= \frac{3}{2} \mathbb{E}_n^T (\mathbf{C}_4 \otimes \mathbb{S}) \mathbb{E}_n, \end{aligned}$$

where $\mathbf{C}_1$, $\mathbf{C}_2$, and $\mathbf{C}_4$ are given by Eqs. (6.29)–(6.31). And $\mathbb{E}$, $\mathbb{E}_n$, $\mathbb{G}_1$, and $\mathbb{G}_2$ are obtained by Eqs. (6.42)–(6.45).

For solving the equation of motion by the Newton–Raphson method, one needs to calculate the tangent stiffness matrix $\mathbb{K}_t$ (Jacobian matrix). We proved that by using the VDQ method, $\mathbb{K}_t$ is obtained easily as follows (see Section 4.4)

$$\mathbb{K}_t(\mathbf{U}) = \frac{d(\mathbb{K}_s \mathbf{U})}{d\mathbf{U}} = \mathbb{K} + \mathbb{N}_1 + \mathbb{N}_2.$$

Properties of $\mathbb{K}_t$ are discussed in Section 4.5.

8. Numerical results and discussion

In this section, we assess the performance of the VDQ method by solving the linear and the nonlinear bending problems of Mindlin plates and comparing the results of VDQ with the results obtained by GDQ and FEM. Two different sets of results are obtained by FEM: using 4-node quadrilateral elements with linear shape functions (FEM-l), and using 9-node quadrilateral elements with quadratic shape functions. These cases are denoted respectively by FEM-l and FEM-q in the tables of the results. The quadratic shape functions are chosen for the purpose of increasing the convergence rate and accuracy of FEM.

The results are given for Mindlin plates with different types of boundary conditions including SSSS, CCCC, CSCS and CFCF (simply-supported, clamped, and free edges are respectively abbreviated to S, C, and F). The corresponding equations for these cases are mentioned in Section 6. Also, the methods of implementing the boundary conditions are discussed in Section 4.6. The considered plates are assumed to be square ($a=b$) with the following properties:

$$\begin{aligned} E &= 1.44 \text{ GPa}, \quad \nu = 0.3, \quad \lambda = \frac{E\nu}{1-\nu^2}, \quad \mu = \frac{E}{2(1+\nu)}, \\ a &= 1 \text{ m}, \quad h = 1/12 \text{ m}, \quad k_s = 5/6. \end{aligned}$$

For the VDQ and GDQ methods, we use the mapped Chebyshev grid points given by Eqs. (a.7) and (a.8) in the Appendix for both edges of the plate. For the 4-node quadrilateral elements (FEM-l), writing $n \times n$ as the number of the grid points indicates a uniform mesh with $(n-1) \times (n-1)$ elements, and for the 9-node quadrilateral elements (FEM-q), it indicates a uniform mesh of $(\frac{n-1}{2}) \times (\frac{n-1}{2})$ elements, where n is chosen to be an odd number.

Table 1

Accuracy versus computational cost for VDQ, GDQ, and FEM in the linear bending analysis of Mindlin plates.

	Method	Number of grid points								
		5 × 5	7 × 7	9 × 9	11 × 11	13 × 13	15 × 15	17 × 17	19 × 19	21 × 21
$\frac{w_{max}}{h}$	VDQ	0.2256	0.2250	0.2254	0.2254	0.2254	0.2254	0.2254	0.2254	0.2254
	GDQ	0.2464	0.2256	0.2253	0.2254	0.2254	0.2254	0.2254	0.2254	0.2254
	FEM-l	0.2177	0.2222	0.2237	0.2243	0.2246	0.2248	0.2250	0.2251	0.2251
	FEM-q	0.2659	0.2279	0.2272	0.2258	0.2258	0.2255	0.2255	0.2254	0.2254
t_w (ms)	VDQ	0.0244	0.1899	0.6031	1.0027	2.3121	4.9291	9.6135	16.9879	30.1398
	GDQ	0.0227	0.4762	2.2457	4.5205	10.8078	21.2133	38.8085	66.9947	124.562
	FEM-l	0.0240	0.0388	0.3281	0.5996	0.8796	1.5996	2.3115	2.9400	3.4185
	FEM-q	0.0221	0.1301	0.3150	0.5784	0.9175	1.3937	2.1975	2.8784	3.2049
nnz	VDQ	405	2425	8085	20,169	42,229	78,585	134,325	215,305	328,149
	GDQ	405	2425	8085	20,169	42,229	78,585	134,325	215,305	328,149
	FEM-l	333	1391	2503	4855	7969	10,595	12,997	18,349	24,501
	FEM-q	425	1981	4645	8547	13,535	19,699	26,881	35,517	45,011

In Table 1, the performance of the VDQ method is compared with that of the GDQ method and FEM in terms of accuracy and computational time. The test problem is the linear bending analysis of a CCCC plate under uniform loading of 100 KN. For the linear analysis of Mindlin plates, we assume $u=v=0$ and reduce the system by taking w , ψ , and φ as the unknowns. The GDQ method is applied to the governing PDEs of plate given in [36]. Our goal is to assess the efficiency of solving the resulting linear system $\mathbb{K}\mathbf{U} = \mathbf{F}$ for the above-mentioned methods. t_w indicates the average value for wall-clock time of solving the linear system $\mathbb{K}\mathbf{U} = \mathbf{F}$ for 100 times, which is reported in milliseconds. Also, the stiffness matrix $\mathbb{K}$ is stored in sparse format and nnz denotes the number of nonzero entries of $\mathbb{K}$ after imposing the boundary conditions. Note that for a $n \times n$ grid points, and after imposing CCCC conditions by the elimination approach, the total entries of $\mathbb{K}$ is $9(n-2)^4$ for all methods. We used MATLAB 2015b linear solver (mldivide, A\b) to solve the linear system $\mathbb{K}\mathbf{U} = \mathbf{F}$, which can take advantage of the properties of $\mathbb{K}$ and execute the optimal solution algorithm. Also, the results are generated on the same computer (Intel Core i7-3632QM @ 2.20 GHz, 8GB of RAM, and Windows 10 64 bit). It is observed that VDQ and GDQ almost have the same rate of convergence and accuracy. Also, regardless of the number of grid points, the corresponding stiffness matrix of VDQ and GDQ has the same level of sparsity (the same nnz). However, the computation time (t_w) of VDQ is almost uniformly less than that of GDQ. The reason is that in contrary to GDQ, VDQ results in a symmetric and positive-definite system (see Section 4.5), which is solved with less computational cost using Cholesky decomposition. Both FEM-l and FEM-q also lead to symmetric and positive-definite systems; but comparing to VDQ, they need much less computation time as the number of grid points increases due to having less sparse matrices. Nevertheless, considering the fast convergence of the VDQ method, it attains the desired level of accuracy in a less computational time among the compared methods (for 9×9 grid points and in 0.6031 ms).

A similar conclusion can be drawn for the performance of VDQ and FEM in a nonlinear bending analysis. At each iteration of Newton method, we solve a linear system with properties similar to what we discussed for Table 1 (see Section 4.5). Also, we observed that if we use the solution of the linear bending problem as the initial guess for the Newton method, it converges in the same number of iterations for both VDQ and FEM. Therefore, according to our observations while preparing the numerical results of the nonlinear bending analysis, albeit FEM generally needs less computational time, for a given level of precision, VDQ achieves the desired solution in a less time with a smaller number of grid points.

In Tables 2–5, the results of VDQ method and FEM are compared for nonlinear bending analysis of plates with different boundary conditions. The applied transverse force is considered to be uniform with the magnitudes of 100α KN, $\alpha = 1, 10, 20, \dots, 50$. For each force amplitude, the solution of linear bending problem is considered as the initial value for the nonlinear solution using the Newton–Raphson method. Also, the tolerance of the results is determined by $\|\mathbb{R}(\mathbf{U})\| \leq 10^{-5}$. The results of FEM using linear shape functions and quadratic shape functions are shown by FEM-l and FEM-q, respectively. At each force amplitude, VDQ, FEM-l, and FEM-q are compared in terms of dimensionless maximum deflection w_{max}/h . It is clearly observed that the convergence rate of the VDQ method is much faster than that of FEM using linear shape functions (FEM-l) for all types of boundary conditions. This is more prominent for large force amplitudes. Comparing to FEM using quadratic shape functions (FEM-q), the convergence rate of the VDQ method is slightly faster except for CFCF boundary conditions, for which the VDQ method has a much faster rate of convergence. It should be noted that in contrast to FEM, VDQ has not the assembly process and also is much easier to implement due to its simple formulation. Fig. 1 is also given for better illustration of the validity of the results.

In FEM analyses of Mindlin plates, we used reduced integration technique to overcome the shear locking problem. In Tables 2–5 and Fig. 1, the converged results of VDQ agree with those obtained by FEM with reduced integration. Thus, we conclude that the VDQ method does not have the shear locking issue. We also checked this for thin plates, for example: In the nonlinear analysis of a CCCC plate with $a=b=1$, $a/h=200$, and uniform loading of 200 N, FEM-q with reduced integration converges to $w_{max}/h=1.4351$ at 35×35 grid points and VDQ converges to $w_{max}/h=1.4352$ at 19×19 grid points.

Table 2Comparison between VDQ and FEM in the nonlinear bending analysis of SSSS Mindlin plates with different force amplitudes α .

α	Method	Number of grid points									
		5 × 5	7 × 7	9 × 9	11 × 11	13 × 13	15 × 15	17 × 17	19 × 19	21 × 21	
w_{max}/h	1	VDQ	0.0660	0.0657	0.0658	0.0658	0.0658	0.0658	0.0658	0.0658	0.0658
		FEM-I	0.0647	0.0653	0.0655	0.0656	0.0657	0.0657	0.0657	0.0657	0.0657
		FEM-q	0.0679	0.0659	0.0659	0.0658	0.0658	0.0658	0.0658	0.0658	0.0658
	10	VDQ	0.4815	0.4865	0.4869	0.4868	0.4868	0.4868	0.4868	0.4868	0.4868
		FEM-I	0.5013	0.4935	0.4906	0.4892	0.4885	0.4880	0.4878	0.4876	0.4874
		FEM-q	0.5038	0.4878	0.4875	0.4869	0.4868	0.4868	0.4868	0.4868	0.4868
	20	VDQ	0.7164	0.7265	0.7268	0.7266	0.7267	0.7267	0.7267	0.7267	0.7267
		FEM-I	0.7650	0.7434	0.7360	0.7326	0.7308	0.7297	0.7290	0.7285	0.7281
		FEM-q	0.7548	0.7285	0.7275	0.7269	0.7268	0.7267	0.7267	0.7267	0.7267
	30	VDQ	0.8729	0.8855	0.8855	0.8853	0.8853	0.8853	0.8853	0.8853	0.8853
		FEM-I	0.9416	0.9093	0.8985	0.8937	0.8911	0.8896	0.8886	0.8879	0.8874
		FEM-q	0.9222	0.8878	0.8863	0.8857	0.8855	0.8854	0.8853	0.8853	0.8853
	40	VDQ	0.9931	1.0072	1.0068	1.0065	1.0065	1.0065	1.0065	1.0065	1.0065
		FEM-I	1.0772	1.0360	1.0227	1.0168	1.0136	1.0117	1.0105	1.0096	1.0090
		FEM-q	1.0507	1.0097	1.0076	1.0069	1.0067	1.0066	1.0066	1.0065	1.0065
	50	VDQ	1.0922	1.1073	1.1063	1.1059	1.1060	1.1059	1.1060	1.1060	1.1060
		FEM-I	1.1886	1.1400	1.1246	1.1177	1.1141	1.1119	1.1105	1.1095	1.1089
		FEM-q	1.1564	1.1097	1.1071	1.1065	1.1061	1.1061	1.1060	1.1060	1.1060

Table 3Comparison between VDQ and FEM in the nonlinear bending analysis of CCCC Mindlin plates with different force amplitudes α .

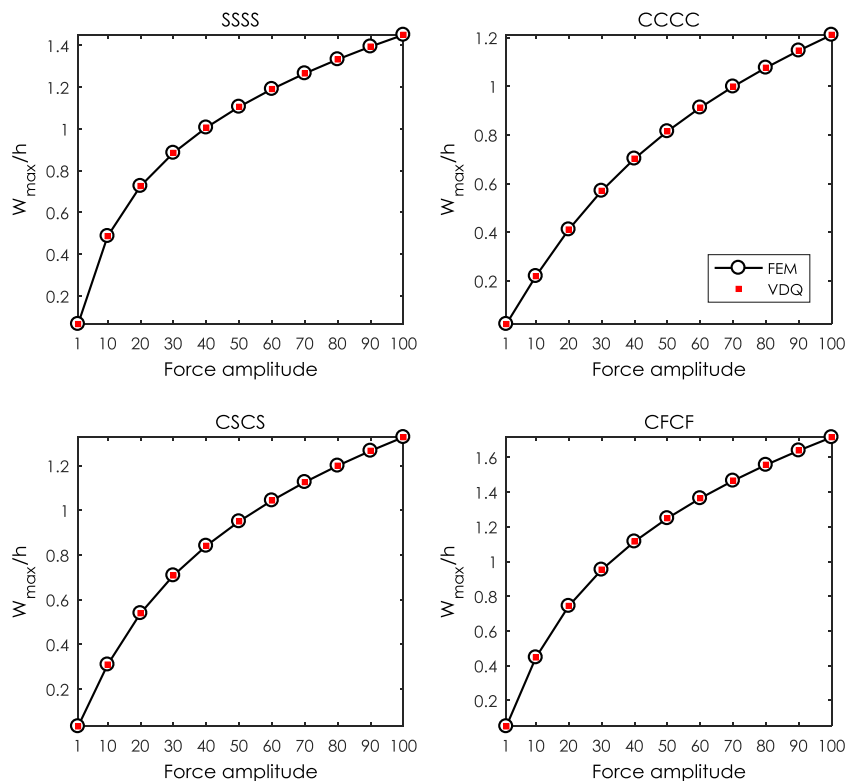
α	Method	Number of grid points									
		5 × 5	7 × 7	9 × 9	11 × 11	13 × 13	15 × 15	17 × 17	19 × 19	21 × 21	
w_{max}/h	1	VDQ	0.0226	0.0225	0.0225	0.0225	0.0225	0.0225	0.0225	0.0225	0.0225
		FEM-I	0.0218	0.0222	0.0224	0.0224	0.0225	0.0225	0.0225	0.0225	0.0225
		FEM-q	0.0266	0.0228	0.0227	0.0226	0.0226	0.0225	0.0225	0.0225	0.0225
	10	VDQ	0.2182	0.2187	0.2194	0.2194	0.2194	0.2194	0.2194	0.2194	0.2194
		FEM-I	0.2144	0.2175	0.2184	0.2187	0.2189	0.2191	0.2191	0.2192	0.2192
		FEM-q	0.2593	0.2216	0.2211	0.2197	0.2197	0.2195	0.2195	0.2194	0.2194
	20	VDQ	0.4040	0.4090	0.4113	0.4111	0.4111	0.4111	0.4111	0.4111	0.4111
		FEM-I	0.4120	0.4122	0.4118	0.4115	0.4114	0.4113	0.4113	0.4112	0.4112
		FEM-q	0.4874	0.4146	0.4139	0.4115	0.4115	0.4112	0.4112	0.4111	0.4111
	30	VDQ	0.5547	0.5662	0.5705	0.5699	0.5699	0.5699	0.5699	0.5699	0.5699
		FEM-I	0.5858	0.5776	0.5742	0.5727	0.5718	0.5713	0.5710	0.5708	0.5706
		FEM-q	0.6776	0.5742	0.5736	0.5704	0.5704	0.5700	0.5700	0.5699	0.5699
	40	VDQ	0.6788	0.6971	0.7031	0.7021	0.7022	0.7021	0.7021	0.7021	0.7021
		FEM-I	0.7367	0.7174	0.7106	0.7074	0.7058	0.7048	0.7042	0.7037	0.7034
		FEM-q	0.8366	0.7070	0.7063	0.7026	0.7027	0.7022	0.7022	0.7021	0.7022
	50	VDQ	0.7838	0.8085	0.8160	0.8144	0.8146	0.8145	0.8145	0.8145	0.8145
		FEM-I	0.8683	0.8372	0.8269	0.8223	0.8198	0.8184	0.8175	0.8168	0.8164
		FEM-q	0.9719	0.8198	0.8190	0.8149	0.8150	0.8145	0.8146	0.8145	0.8145

Table 4Comparison between VDQ and FEM in the nonlinear bending analysis of CSCS Mindlin plates with different force amplitudes α .

α	Method	Number of grid points									
		5 × 5	7 × 7	9 × 9	11 × 11	13 × 13	15 × 15	17 × 17	19 × 19	21 × 21	
$w_{\max}/h$	1	VDQ	0.0338	0.0333	0.0333	0.0333	0.0333	0.0333	0.0333	0.0333	0.0333
		FEM-I	0.0310	0.0324	0.0328	0.0330	0.0331	0.0332	0.0332	0.0332	0.0333
		FEM-q	0.0361	0.0334	0.0335	0.0333	0.0334	0.0333	0.0333	0.0333	0.0333
	10	VDQ	0.3103	0.3087	0.3098	0.3097	0.3097	0.3097	0.3097	0.3097	0.3097
		FEM-I	0.2975	0.3052	0.3073	0.3082	0.3087	0.3089	0.3091	0.3092	0.3093
		FEM-q	0.3385	0.3108	0.3108	0.3098	0.3099	0.3097	0.3097	0.3097	0.3097
	20	VDQ	0.5327	0.5369	0.5397	0.5395	0.5395	0.5395	0.5395	0.5395	0.5395
		FEM-I	0.5418	0.5417	0.5408	0.5404	0.5401	0.5399	0.5398	0.5398	0.5397
		FEM-q	0.5964	0.5415	0.5413	0.5396	0.5397	0.5395	0.5395	0.5395	0.5395
	30	VDQ	0.6943	0.7048	0.7090	0.7085	0.7085	0.7085	0.7085	0.7085	0.7085
		FEM-I	0.7335	0.7199	0.7148	0.7125	0.7113	0.7105	0.7100	0.7097	0.7095
		FEM-q	0.7889	0.7114	0.7108	0.7086	0.7088	0.7085	0.7085	0.7085	0.7085
	40	VDQ	0.8210	0.8368	0.8419	0.8409	0.8411	0.8410	0.8410	0.8410	0.8410
		FEM-I	0.8881	0.8607	0.8517	0.8478	0.8457	0.8444	0.8436	0.8431	0.8427
		FEM-q	0.9408	0.8448	0.8437	0.8412	0.8413	0.8410	0.8410	0.8410	0.8410
	50	VDQ	0.9259	0.9460	0.9517	0.9503	0.9505	0.9504	0.9504	0.9504	0.9504
		FEM-I	1.0172	0.9772	0.9649	0.9595	0.9567	0.9550	0.9539	0.9531	0.9526
		FEM-q	1.0664	0.9549	0.9533	0.9506	0.9507	0.9504	0.9504	0.9504	0.9504

Table 5Comparison between VDQ and FEM in the nonlinear bending analysis of CFCF Mindlin plates with different force amplitudes α .

α	Method	Number of grid points									
		5 × 5	7 × 7	9 × 9	11 × 11	13 × 13	15 × 15	17 × 17	19 × 19	21 × 21	
w_{max}/h	1	VDQ	0.0508	0.0503	0.0505	0.0504	0.0504	0.0504	0.0504	0.0504	0.0504
		FEM-I	0.0382	0.0449	0.0472	0.0484	0.0490	0.0493	0.0496	0.0497	0.0499
		FEM-q	0.0513	0.0497	0.0503	0.0502	0.0503	0.0503	0.0503	0.0503	0.0504
	10	VDQ	0.4429	0.4430	0.4487	0.4479	0.4477	0.4476	0.4475	0.4475	0.4475
		FEM-I	0.3646	0.4130	0.4282	0.4353	0.4390	0.4412	0.4427	0.4437	0.4444
		FEM-q	0.4660	0.4421	0.4468	0.4460	0.4469	0.4468	0.4471	0.4471	0.4472
	20	VDQ	0.7243	0.7332	0.7462	0.7446	0.7444	0.7442	0.7442	0.7441	0.7441
		FEM-I	0.6581	0.7114	0.7259	0.7327	0.7361	0.7382	0.7395	0.7405	0.7411
		FEM-q	0.7905	0.7362	0.7433	0.7417	0.7430	0.7430	0.7434	0.7435	0.7437
	30	VDQ	0.9194	0.9388	0.9560	0.9535	0.9534	0.9532	0.9531	0.9531	0.9531
		FEM-I	0.8850	0.9282	0.9388	0.9442	0.9467	0.9484	0.9494	0.9501	0.9507
		FEM-q	1.0215	0.9439	0.9523	0.9502	0.9517	0.9517	0.9522	0.9523	0.9525
	40	VDQ	1.0701	1.0995	1.1192	1.1158	1.1158	1.1155	1.1155	1.1155	1.1155
		FEM-I	1.0667	1.0978	1.1047	1.1088	1.1105	1.1118	1.1126	1.1131	1.1135
		FEM-q	1.2006	1.1055	1.1147	1.1122	1.1139	1.1138	1.1145	1.1145	1.1148
	50	VDQ	1.1942	1.2328	1.2542	1.2497	1.2499	1.2496	1.2496	1.2495	1.2495
		FEM-I	1.2181	1.2379	1.2416	1.2447	1.2458	1.2467	1.2473	1.2477	1.2480
		FEM-q	1.3479	1.2390	1.2487	1.2460	1.2478	1.2478	1.2485	1.2485	1.2488

**Figure 1.** Converged VDQ and FEM results for nonlinear bending analysis of SSSS, CCCC, CSCS, and CFCF Mindlin plates. FEM results are generated using quadratic shape functions.

In Tables 6 and 7, we compare the performance of VDQ and FEM for nonlinear bending analysis of plates with SSSS and CCCC boundary conditions under non-uniform transverse loads. We consider three cases of distributed loadings as follows:

$$f_1(x_1, x_2) = \alpha x_2, \quad f_2(x_1, x_2) = \alpha \sin(2\pi x_1) \cos(2\pi x_2), \quad f_3(x_1, x_2) = \begin{cases} \alpha, & 0 \leq x_1 \leq \frac{a}{2}, \\ 0, & \frac{a}{2} < x_1 \leq a, \end{cases}$$

where $\alpha = 5000$ KN. We use the following norm to compare the deflection $w(x_1, x_2)$ over the entire plate domain

$$\|w\| = \frac{1}{abh} \left(\int_0^a \int_0^b (w(x_1, x_2))^2 dx_2 dx_1 \right)^{\frac{1}{2}}.$$

Table 6

Comparison of VDQ and FEM in the nonlinear bending analysis of SSSS Mindlin plates with different non-uniform loadings.

Transverse force	Method	Number of grid points								
		7 × 7	9 × 9	11 × 11	13 × 13	15 × 15	17 × 17	19 × 19	21 × 21	23 × 23
f_1	VDQ	0.4260	0.4258	0.4258	0.4258	0.4258	0.4258	0.4258	0.4258	0.4258
	FEM-l	0.4125	0.4184	0.4211	0.4225	0.4234	0.4239	0.4243	0.4246	0.4248
	FEM-q	0.4247	0.4254	0.4256	0.4257	0.4257	0.4257	0.4257	0.4257	0.4258
f_2	VDQ	0.0741	0.0747	0.0747	0.0747	0.0747	0.0747	0.0747	0.0747	0.0747
	FEM-l	0.0606	0.0666	0.0694	0.0710	0.0720	0.0726	0.0731	0.0734	0.0736
	FEM-q	0.0707	0.0733	0.0741	0.0744	0.0745	0.0746	0.0746	0.0747	0.0747
f_3	VDQ	0.4225	0.4215	0.4212	0.4210	0.4209	0.4208	0.4207	0.4207	0.4207
	FEM-l	0.4082	0.4137	0.4162	0.4176	0.4184	0.4189	0.4193	0.4195	0.4197
	FEM-q	0.4664	0.4202	0.4494	0.4205	0.4415	0.4206	0.4370	0.4206	0.4341

Table 7

Comparison of VDQ and FEM in the nonlinear bending analysis of CCCC Mindlin plates with different non-uniform loadings.

Transverse force	Method	Number of grid points								
		7 × 7	9 × 9	11 × 11	13 × 13	15 × 15	17 × 17	19 × 19	21 × 21	23 × 23
f_1	VDQ	0.2217	0.2217	0.2217	0.2217	0.2217	0.2217	0.2217	0.2217	0.2217
	FEM-l	0.2039	0.2117	0.2153	0.2173	0.2185	0.2192	0.2197	0.2201	0.2204
	FEM-q	0.2195	0.2210	0.2214	0.2215	0.2216	0.2216	0.2217	0.2217	0.2217
f_2	VDQ	0.0472	0.0475	0.0475	0.0475	0.0475	0.0475	0.0475	0.0475	0.0475
	FEM-l	0.0334	0.0393	0.0422	0.0438	0.0448	0.0454	0.0458	0.0462	0.0464
	FEM-q	0.0420	0.0457	0.0468	0.0472	0.0473	0.0474	0.0475	0.0475	0.0475
f_3	VDQ	0.2238	0.2241	0.2243	0.2244	0.2245	0.2245	0.2245	0.2246	0.2246
	FEM-l	0.2060	0.2142	0.2179	0.2200	0.2212	0.2220	0.2226	0.2230	0.2233
	FEM-q	0.2654	0.2237	0.2509	0.2244	0.2438	0.2246	0.2396	0.2246	0.2370

Table 8

Comparison of VDQ with FEM on obtaining the stress resultants in the nonlinear bending analysis of SSSS Mindlin plates.

Resultant	Method	Number of grid points								
		5 × 5	7 × 7	9 × 9	11 × 11	13 × 13	15 × 15	17 × 17	19 × 19	21 × 21
$\ N_{11}\ , \ N_{22}\ $	VDQ	22.5540	22.8703	22.8638	22.8643	22.8644	22.8644	22.8644	22.8644	22.8644
	FEM-l	21.4021	22.2463	22.5226	22.6473	22.7142	22.7543	22.7803	22.7980	22.8107
	FEM-q	22.8786	22.8745	22.8670	22.8652	22.8647	22.8645	22.8645	22.8644	22.8644
$\ N_{12}\ $	VDQ	0.7774	0.8911	0.9024	0.9019	0.9019	0.9018	0.9018	0.9018	0.9018
	FEM-l	2.6959	1.7883	1.4333	1.2561	1.1545	1.0909	1.0484	1.0188	0.9972
	FEM-q	1.4226	0.9497	0.9001	0.8991	0.9004	0.9011	0.9015	0.9016	0.9017
$\ M_{11}\ , \ M_{22}\ $	VDQ	0.4728	0.4784	0.4784	0.4784	0.4784	0.4784	0.4784	0.4784	0.4784
	FEM-l	0.4642	0.4725	0.4752	0.4764	0.4770	0.4774	0.4776	0.4778	0.4779
	FEM-q	0.4660	0.4756	0.4775	0.4780	0.4782	0.4783	0.4783	0.4783	0.4783
$\ M_{12}\ $	VDQ	0.2478	0.2519	0.2519	0.2519	0.2519	0.2519	0.2519	0.2519	0.2519
	FEM-l	0.2477	0.2502	0.2510	0.2513	0.2515	0.2516	0.2517	0.2517	0.2518
	FEM-q	0.2485	0.2510	0.2516	0.2518	0.2519	0.2519	0.2519	0.2519	0.2519
$\ Q_1\ , \ Q_2\ $	VDQ	2.9751	2.5978	2.5935	2.5930	2.5929	2.5928	2.5928	2.5928	2.5928
	FEM-l	13.4283	9.3095	7.2250	5.9975	5.2042	4.6588	4.2666	3.9749	3.7521
	FEM-q	4.0023	3.3109	2.9537	2.7778	2.6936	2.6514	2.6288	2.6160	2.6084

Since, in the VDQ method, the numerical integration is performed globally using the data at the grid points, a non-continuous function like f_3 causes inaccuracy. To overcome this issue, we estimate f_3 using the error function as

$$f_3 \approx \frac{\alpha}{2} - \frac{\alpha}{2} \operatorname{erf}\left(k\left(x_1 - \frac{a}{2}\right)\right), \quad \operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-t^2} dt,$$

where k is chosen as 100. For both SSSS and CCCC boundary conditions and for all cases of non-uniform loadings considered above, one can see that VDQ is more accurate than FEM-l and FEM-q.

We compare the accuracy of VDQ with that of FEM in approximating the stress resultants, in Tables 8 and 9 for plates with SSSS and CCCC boundary conditions, respectively. The transverse force is uniform with the magnitude of $\alpha = 5000$ KN and results are for the nonlinear bending analysis. Consistent to the notation that we used in this work, the stress resultants can be obtained as follows:

$$\begin{bmatrix} N_{11} \\ N_{22} \\ N_{12} \end{bmatrix} = h \mathbf{C}_{in} \left(\mathbf{E} \begin{bmatrix} u \\ v \end{bmatrix} + (\mathbf{G}_1 w) \circ (\mathbf{G}_2 w) \right), \quad \begin{bmatrix} M_{11} \\ M_{22} \\ M_{12} \end{bmatrix} = \frac{h^3}{12} \mathbf{C}_{in} \mathbf{E} \begin{bmatrix} \psi \\ \phi \end{bmatrix}, \quad \begin{bmatrix} Q_1 \\ Q_2 \end{bmatrix} = h \mathbf{C}_{out} \mathbf{E}_s \begin{bmatrix} w \\ \psi \\ \phi \end{bmatrix},$$

Table 9

Comparison of VDQ with FEM on obtaining the stress resultants in the nonlinear bending analysis of CCCC Mindlin plates.

Resultant	Method	Number of grid points								
		5 × 5	7 × 7	9 × 9	11 × 11	13 × 13	15 × 15	17 × 17	19 × 19	21 × 21
$\ N_{11}\ , \ N_{22}\ $	VDQ	9.9673	10.0043	10.0982	10.0969	10.0961	10.0957	10.0956	10.0956	10.0956
	FEM-l	8.0004	9.2096	9.6035	9.7826	9.8790	9.9368	9.9742	9.9997	10.0180
	FEM-q	10.0071	10.0233	10.0788	10.0928	10.0962	10.0970	10.0971	10.0969	10.0967
$\ N_{12}\ $	VDQ	0.8059	1.1935	1.1340	1.1229	1.1224	1.1224	1.1224	1.1224	1.1224
	FEM-l	2.0121	1.6469	1.4414	1.3355	1.2742	1.2358	1.2102	1.1923	1.1793
	FEM-q	1.8814	1.4174	1.2557	1.1857	1.1549	1.1404	1.1331	1.1292	1.1269
$\ M_{11}\ , \ M_{22}\ $	VDQ	0.4811	0.5343	0.5297	0.5300	0.5300	0.5300	0.5300	0.5300	0.5300
	FEM-l	0.5032	0.5166	0.5225	0.5252	0.5267	0.5276	0.5281	0.5285	0.5288
	FEM-q	0.5155	0.5255	0.5278	0.5288	0.5293	0.5295	0.5297	0.5298	0.5298
$\ M_{12}\ $	VDQ	0.1336	0.1512	0.1513	0.1516	0.1516	0.1516	0.1516	0.1516	0.1516
	FEM-l	0.1385	0.1474	0.1494	0.1502	0.1507	0.1509	0.1511	0.1512	0.1513
	FEM-q	0.1574	0.1522	0.1518	0.1517	0.1516	0.1516	0.1516	0.1516	0.1516
$\ Q_1\ , \ Q_2\ $	VDQ	6.0988	5.8132	5.8162	5.8107	5.8099	5.8100	5.8102	5.8102	5.8102
	FEM-l	18.0049	12.9177	10.4874	9.1089	8.2521	7.6850	7.2915	7.0080	6.7975
	FEM-q	16.3633	10.6271	8.0000	6.8834	6.3808	6.1367	6.0091	5.9378	5.8956

$$\mathbf{E} = \begin{bmatrix} \partial/\partial x_1 & 0 \\ 0 & \partial/\partial x_2 \\ \partial/\partial x_2 & \partial/\partial x_1 \end{bmatrix}, \mathbf{E}_s = \begin{bmatrix} \partial/\partial x_1 & 1 & 0 \\ \partial/\partial x_2 & 0 & 1 \end{bmatrix}, \mathbf{G}_1 = \begin{bmatrix} \partial/\partial x_1 \\ \partial/\partial x_2 \\ \partial/\partial x_1 \end{bmatrix}, \mathbf{G}_2 = \frac{1}{2} \begin{bmatrix} \partial/\partial x_1 \\ \partial/\partial x_2 \\ 2(\partial/\partial x_2) \end{bmatrix},$$

where $\mathbf{C}_{in}$ and $\mathbf{C}_{out}$ are given by Eqs. (6.22) and (6.23). Moreover, to compare the stress resultants, we use the following norm for a stress resultant $f(x_1, x_2)$ defined in the plate domain

$$\|f\| = \frac{1}{ab\beta} \left(\int_0^a \int_0^b (f(x_1, x_2))^2 dx_2 dx_1 \right)^{\frac{1}{2}},$$

in which $\beta = 100$ KN. Also, we have checked the numerical results by the fact that, due to symmetry, $\|N_{11}\| = \|N_{22}\|$, $\|M_{11}\| = \|M_{22}\|$, and $\|Q_1\| = \|Q_2\|$. Among the compared methods, we can see that the VDQ method have a better accuracy and a faster convergence rate for approximating all stress resultants.

9. Conclusion

A novel numerical method in the area of computational mechanics called Variational Differential Quadrature (VDQ) was proposed in this paper. This technique is applied to the weak form of the equations. In the VDQ method, by introducing an efficient matrix formulation and using a precise integral operator, the functional of energy is discretized and written as a quadratic form, from which the discretized governing equations are derived readily in a direct way with no need for the analytical derivation of partial differential equations of the strong form. The following features can be highlighted for the VDQ method:

- The approach of the VDQ method is general so that it can be adopted for any engineering structure, which is modeled by energy methods and using the weak form of the equations. In this work, the VDQ method was formulated for the nonlinear elasticity and it was simplified further based on the Mindlin plate theory and the Timoshenko beam theory. Similarly, it can be applied to other engineering structures.
- The VDQ method has a compact and an explicit matrix formulation, which is easy to implement.
- From a numerical point of view, the convergence rate of VDQ is faster than that of FEM in the nonlinear bending analysis of plates. Also, comparing to GDQ, VDQ needs less computational cost due to having symmetric positive-definite matrices.

Hence, the VDQ method, due to its simplicity and accuracy can be very promising for numerical analysis of structures in engineering. In the implementation of the VDQ method, we used GDQ differential operators to approximate the derivatives in strain tensor. Also, the integral operator was formulated based on the GDQ relations. Although using GDQ differential operators leads to the precision of the VDQ method, it also induces some limitations. For example, it would be difficult to use the VDQ method for domains with non-coordinate aligned boundary conditions, discontinuities, and complex geometries. Nonetheless, the VDQ method offers an effective formulation which provides a framework to directly discretize the energy functionals; which is independent of the types of the differential and integral operators used in the method. In future works, we aim to use the same formulation with different differential and integral operators such as moving least-square differential quadrature (MLSDQ) operators [37] to overcome the limitations mentioned above. We also aim to extend the application of current work by using less sparse operators of FDM, adaptive mesh refinements, and multigrid techniques.

Appendix

a. Differential operator

The GDQ differential operators are more suitable than the other ones, because they can be readily constructed and have a negligible error. Let $\mathbf{D}_x^{(r)}$ be the differential operator of r th derivatives along the x direction. Its components are defined as [15]

$$D_{xij}^{(r)} = \begin{cases} \delta_{ij} & r = 0, \\ \frac{P_i}{(x_i - x_j)P_j} & i \neq j, \quad r = 1, \\ r \left(D_{xij}^{(1)} D_{xii}^{(r-1)} - \frac{D_{xij}^{(r-1)}}{x_i - x_j} \right) & i \neq j, \quad r = 2, 3, \dots, n-1, \\ -\sum_{j=1, j \neq i}^n D_{xij}^{(r)} & i = j, \quad r = 1, 2, 3, \dots, n-1, \end{cases} \quad (\text{a.1})$$

for $i, j = 1, 2, \dots, n$, where n is the number of grid points, δ_{ij} is the Kronecker delta, and the constant P_j is given by

$$P_j = \prod_{\substack{i=1 \\ i \neq j}}^n (x_j - x_i). \quad (\text{a.2})$$

For a function $f(x)$, by discretizing along the x direction and calculating the function values at grid points, the following vector can be constructed

$$\mathbf{F} = [f(x_1) \quad f(x_2) \quad \dots \quad f(x_n)]^T, \quad (\text{a.3})$$

where $f(x_i)$ is the function value at the grid point x_i . Now, if one write

$$f^{(r)}(x_i) = \left. \frac{d^r f(x)}{dx^r} \right|_{x=x_i}, \quad (\text{a.4})$$

it is possible to represent the vector of nodal values of r -th derivative of $f(x)$ at grid points as follows

$$\mathbf{F}^{(r)} = [f^{(r)}(x_1) \quad f^{(r)}(x_2) \quad \dots \quad f^{(r)}(x_n)]^T. \quad (\text{a.5})$$

Using GDQ differential operators, the above relation can be approximated as

$$\mathbf{F}^{(r)} = \mathbf{D}_x^{(r)} \mathbf{F}. \quad (\text{a.6})$$

To minimize the error of differential operator, the set of grid points must be carefully selected. It can be proven for GDQ differential operators that the approximation error is minimum when the Chebyshev grids are selected for the distributions of points within $[-1, 1]$. In this work, the grid points are selected to be the extrema of the Chebyshev polynomial of degree $n-1$, (known as Chebyshev-Gauss-Lobatto grid) which are located at

$$\hat{x}_i = -\cos \frac{i-1}{n-1} \pi, \quad i = 1, 2, \dots, n, \quad -1 \leq \hat{x}_i \leq 1. \quad (\text{a.7})$$

Accordingly, the grid can be mapped onto $[a, b]$ as

$$x_i = \frac{(b-a)}{2} \hat{x}_i + \frac{a+b}{2}, \quad a \leq x_i \leq b. \quad (\text{a.8})$$

b. Integral operator

Since the integration is performed on the entire interval, using an inappropriate integral operator can lead to error propagation. The selected integral operator must be proportional to its DQ differential counterpart so that the integral values are calculated at the same points on which differentiation is calculated. To this end, Taylor series is employed to evaluate the function $f(x)$ in a neighborhood of x_i as

$$f(x) = \sum_{r=0}^{\infty} \frac{(x - x_i)^r}{r!} f^{(r)}(x_i). \quad (\text{b.1})$$

Assume that $I_i = [\frac{x_{i-1} + x_i}{2}, \frac{x_i + x_{i+1}}{2}]$ is a subset of the interval of convergence of the above power series implying that convergence is uniform for all $x \in I_i$. Then, integrating above equation over I_i results in

$$\int_{\frac{x_{i-1} + x_i}{2}}^{\frac{x_i + x_{i+1}}{2}} f(x) dx = \sum_{r=0}^{\infty} \frac{(x_{i+1} - x_i)^{r+1} - (x_{i-1} - x_i)^{r+1}}{2^{r+1} (r+1)!} f^{(r)}(x_i). \quad (\text{b.2})$$

Now, $[x_1, x_n]$ is divided into n subintervals as follows

$$[x_1, x_n] = \left[x_1, \frac{x_1 + x_2}{2} \right] \cup \left[\frac{x_1 + x_2}{2}, \frac{x_2 + x_3}{2} \right] \cup \dots \cup \left[\frac{x_{n-2} + x_{n-1}}{2}, \frac{x_{n-1} + x_n}{2} \right] \cup \left[\frac{x_{n-1} + x_n}{2}, x_n \right]. \quad (\text{b.3})$$

Then, one can introduce the following relation

$$\int_{x_1}^{x_n} f(x) dx = \int_{x_1}^{\frac{x_1 + x_2}{2}} f(x) dx + \sum_{i=2}^{n-1} \int_{\frac{x_{i-1} + x_i}{2}}^{\frac{x_i + x_{i+1}}{2}} f(x) dx + \int_{\frac{x_{n-1} + x_n}{2}}^{x_n} f(x) dx. \quad (\text{b.4})$$

By calculating the first and last integrals of this relation through the integration of Taylor series in Eq. (b.1) and using Eq. (b.2), the integral of Eq. (b.4) can be calculated as follows

$$\int_{x_1}^{x_n} f(x) dx = \sum_{r=0}^{\infty} \left(\frac{(x_2 - x_1)^{r+1}}{2^{r+1}(r+1)!} f^{(r)}(x_1) + \sum_{i=2}^{n-1} \frac{(x_{i+1} - x_i)^{r+1} - (x_{i-1} - x_i)^{r+1}}{2^{r+1}(r+1)!} f^{(r)}(x_i) - \frac{(x_{n-1} - x_n)^{r+1}}{2^{r+1}(r+1)!} f^{(r)}(x_n) \right). \quad (\text{b.5})$$

If for $i=2, 3, \dots, n-1$, $\tilde{\mathbf{X}}$ be defined as

$$\tilde{\mathbf{X}}^{(r)} = \left[\frac{(x_2 - x_1)^{r+1}}{2^{r+1}(r+1)!} \quad \dots \quad \frac{(x_{i+1} - x_i)^{r+1} - (x_{i-1} - x_i)^{r+1}}{2^{r+1}(r+1)!} \quad \dots \quad - \frac{(x_{n-1} - x_n)^{r+1}}{2^{r+1}(r+1)!} \right], \quad (\text{b.6})$$

Eq. (b.5) can be given by

$$\int_{x_1}^{x_n} f(x) dx = \sum_{r=0}^{\infty} \tilde{\mathbf{X}}^{(r)} \mathbf{F}^{(r)}, \quad (\text{b.7})$$

where $\mathbf{F}^{(r)}$ is given by Eq. (a.5). In accordance with Eq. (a.1), the above relation can be approximated as

$$\int_{x_1}^{x_n} f(x) dx = \left(\sum_{r=0}^{n-1} \tilde{\mathbf{X}}^{(r)} \mathbf{D}_x^{(r)} \right) \mathbf{F} = \tilde{\mathbf{S}}_x \mathbf{F}, \quad (\text{b.8})$$

in which $\tilde{\mathbf{S}}_x$ is the integral operator along the x direction defined by

$$\tilde{\mathbf{S}}_x = \sum_{r=0}^{n-1} \tilde{\mathbf{X}}^{(r)} \mathbf{D}_x^{(r)}. \quad (\text{b.9})$$

c. Generalization to N-dimensional space

Let the number of grid points along the x_i direction in a N -dimensional space be equal to n_i . For the function $f(x_1, \dots, x_N)$, the values at the grid points $F_{i,j,\dots,k}$ are defined as

$$F_{i,j,\dots,k} = f(x_{1i}, x_{2j}, \dots, x_{Nk}), \quad \begin{cases} i = 1, \dots, n_1, \\ j = 1, \dots, n_2, \\ \vdots \\ k = 1, \dots, n_N. \end{cases} \quad (\text{c.1})$$

Using these grid values, one can construct the following $n_1 n_2 \dots n_N \times 1$ vector

$$\mathbb{F} = [F_{1,1,1,\dots,1} \quad \dots \quad F_{n_1,1,1,\dots,1} \quad F_{1,2,1,\dots,1} \quad \dots \quad F_{n_1,n_2,1,\dots,1} \quad F_{1,1,2,\dots,1} \quad \dots \quad F_{n_1,n_2,n_3,\dots,n_N}]^T. \quad (\text{c.2})$$

Using the GDQ differential operators, the partial differentiation in the N -dimensional space can be defined as

$$\frac{\partial^{r_1+r_2+\dots+r_N} f(x_1, x_2, \dots, x_N)}{\partial x_1^{r_1} \partial x_2^{r_2} \dots \partial x_N^{r_N}} = (\mathbf{D}_{x_N}^{(r_N)} \otimes \mathbf{D}_{x_{N-1}}^{(r_{N-1})} \otimes \dots \otimes \mathbf{D}_{x_2}^{(r_2)} \otimes \mathbf{D}_{x_1}^{(r_1)}) \mathbb{F} = \mathbb{D}_{x_1 x_2 \dots x_N}^{(r_1, r_2, \dots, r_N)} \mathbb{F}. \quad (\text{c.3})$$

Similarly, one can also generalize the integral operator to the N -dimensional space as follows

$$\int_{x_{11}}^{x_{1n_1}} \int_{x_{21}}^{x_{2n_2}} \dots \int_{x_{N1}}^{x_{Nn_N}} f(x_1, x_2, \dots, x_N) dx_1 dx_2 \dots dx_N = (\tilde{\mathbf{S}}_{x_N} \otimes \tilde{\mathbf{S}}_{x_{N-1}} \otimes \dots \otimes \tilde{\mathbf{S}}_{x_2} \otimes \tilde{\mathbf{S}}_{x_1}) \mathbb{F} = \tilde{\mathbf{S}}_{x_1 x_2 \dots x_N} \mathbb{F}, \quad (\text{c.4})$$

where the integral operator $\tilde{\mathbf{S}}_{x_1 x_2 \dots x_N}$ is defined as

$$\tilde{\mathbf{S}}_{x_1 x_2 \dots x_N} = \tilde{\mathbf{S}}_{x_N} \otimes \tilde{\mathbf{S}}_{x_{N-1}} \otimes \dots \otimes \tilde{\mathbf{S}}_{x_2} \otimes \tilde{\mathbf{S}}_{x_1}. \quad (\text{c.5})$$

d. Grid points in 3-dimensional space

Two different approaches with the same results can be employed so as to generate grid points in $\mathbb{R}^3$. In the first approach, it is possible to separately discretize the component x_i of $\mathbf{x} = [x_1 \ x_2 \ x_3]$ on n_i points and to construct the vector $\mathbf{X}_i$. Thus, by denoting the value of the component x_i at the j -th point as $x_{i,j}$, the vector $\mathbf{X}_i$ is constructed as follows

$$\mathbf{X}_i = \begin{bmatrix} x_{i,1} \\ x_{i,2} \\ \vdots \\ x_{i,n_i} \end{bmatrix}_{n_i \times 1}, \quad i = 1, 2, 3. \quad (\text{d.1})$$

The coordinates of the grid points in the 3-dimensional space can be represented by a $n \times 3$ matrix ($n = n_1 n_2 n_3$) as

$$\mathbb{X} = [\mathbf{1}_{n_3 \times 1} \otimes \mathbf{1}_{n_2 \times 1} \otimes \mathbf{X}_1 \quad \mathbf{1}_{n_3 \times 1} \otimes \mathbf{X}_2 \otimes \mathbf{1}_{n_1 \times 1} \quad \mathbf{X}_3 \otimes \mathbf{1}_{n_2 \times 1} \otimes \mathbf{1}_{n_1 \times 1}], \quad (\text{d.2})$$

where $\mathbf{1}_{n_i \times 1}$ is an $n_i \times 1$ vector whose elements are all one.

In the second approach, one can count indices for three dimensions and obtain the matrix given by Eq. (d.2). Hence

$$\mathbb{X} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{bmatrix}, \quad \mathbf{x}_l = [x_{1,i} \quad x_{2,j} \quad x_{3,k}], \quad (\text{d.3})$$

where $n = n_1 n_2 n_3$ and the index l is calculated by

$$l = n_1((n_2(k-1) + j) - 1) + i, \quad \begin{cases} i = 1, 2, \dots, n_1, \\ j = 1, 2, \dots, n_2, \\ k = 1, 2, \dots, n_3. \end{cases} \quad (\text{d.4})$$

Note that if we set $N=3$ in Eq. (c.2), it becomes the discretized form of an arbitrary function on the grid points $\mathbb{X}$.

e. Error analysis

In this section, we discuss an outline for the error analysis of the VDQ method. Error in the discretization of a functional using the VDQ method is caused by numerical evaluation of the integral and approximation of the derivatives in the integrand. For approximating the derivatives, we have used GDQ differential operators given by Eq. (a.1), and for approximating the integral, we have introduced an integral operator in Section b of the Appendix. Herein, we discuss the error of those approximations. Moreover, we estimate the error for the VDQ method in an example; the same approach can be applied to any other problem.

From the Shu's work [15], the error for approximating a function $f(x)$ using Lagrange polynomials of degree $n-1$ can be obtained by

$$E[f(x)] = \frac{f^{(n)}(\xi)M(x)}{n!}, \quad (\text{e.1})$$

where $M(x) = (x-x_1)(x-x_2)\dots(x-x_n)$, $f^{(n)}$ denotes the n th derivative of $f(x)$, and ξ is a point between x_1 and x_n , which are the first and the last points on the grid. In general, ξ is a function of x . Since GDQ differential operators is derived using the Lagrange polynomials, the error for the r th order derivative of $f(x)$ can be calculated by

$$E_D^{(r)}[f(x)] = \frac{1}{n!} \frac{d^r(f^{(n)}(\xi)M(x))}{dx^r}. \quad (\text{e.2})$$

But since ξ is a function of x , working with the above formula is difficult. To overcome this, Shu [15] assumed that n th order derivative of $f(x)$ is bounded by a constant c , that is $|f^{(n)}(x)| \leq c$, and showed that

$$|E_D^{(r)}[f(x)]| \leq \frac{ch^{n-r}}{(n-r)!}, \quad r = 1, 2, \dots, n-1, \quad (\text{e.3})$$

where h is the length of the interval containing all the grid points, that is $h = |x_n - x_1|$. For detailed derivation of Eqs. (e.1) and (e.3) see Section 2.4 in [15].

We define the error for the integral operator, Eq. (b.9), as

$$E_S[f] = \int_{x_1}^{x_n} f(x)dx - \tilde{\mathbf{S}}_x \mathbf{F}. \quad (\text{e.4})$$

Moreover, we write Eq. (b.7) as

$$\int_{x_1}^{x_n} f(x) dx = \sum_{r=0}^{n-1} \tilde{\mathbf{X}}^{(r)} \mathbf{F}^{(r)} + R[f], \quad (\text{e.5})$$

where $R[f]$ is the integral of remainder of the Taylor series. Using GDQ differential operators given by Eq. (a.1), we get

$$\int_{x_1}^{x_n} f(x) dx = \tilde{\mathbf{X}}^{(0)} \mathbf{F} + \sum_{r=1}^{n-1} \tilde{\mathbf{X}}^{(r)} (\mathbf{D}_x^{(r)} \mathbf{F} + \mathbf{E}_D^{(r)}) + R[f], \quad (\text{e.6})$$

where $\mathbf{E}_D^{(r)} = [E_D^{(r)}[f(x_1)] \cdots E_D^{(r)}[f(x_n)]]^T$ is a vector containing the errors of approximating the derivatives at the grid points. According to Eqs. (b.9) and (e.6), the error of integration becomes

$$E_S[f] = \sum_{r=1}^{n-1} \tilde{\mathbf{X}}^{(r)} \mathbf{E}_D^{(r)} + R[f]. \quad (\text{e.7})$$

First, we calculate $R[f]$. For the grid points x_i , $i=1, 2, \dots, n$, define $I_i = [a_i, b_i]$, where $a_i = \frac{x_{i-1} + x_i}{2}$ and $b_i = \frac{x_i + x_{i+1}}{2}$ with the exceptions $a_1 = x_1$ and $b_n = x_n$. Let U be an open interval in $\mathbb{R}$ containing the closed interval $[x_1, x_n]$ and let $f: U \rightarrow \mathbb{R}$ be n times differentiable. Then, using Taylor's theorem, for any point $x \in I_i$, we have the following remainder

$$r_i(x) = f(x) - \sum_{r=0}^{n-1} \frac{(x - x_i)^r}{r!} f^{(r)}(x_i) = \frac{(x - x_i)^n}{n!} f^{(n)}(\xi_i), \quad (\text{e.8})$$

where ξ_i is some number between x and x_i and $\xi_i = x_i$, whenever $x = x_i$. Now, similar to Eq. (b.4), by taking the integral of the reminder r_i over I_i for $i=1, 2, \dots, n$ and then adding the results, we get

$$R[f] = \sum_{i=1}^n \int_{a_i}^{b_i} r_i(x) dx = \frac{1}{n!} \sum_{i=1}^n \int_{a_i}^{b_i} (x - x_i)^n f^{(n)}(\xi_i) dx. \quad (\text{e.9})$$

Note that ξ_i depends on x . Assuming that $|f^{(n)}(x)| \leq c$, we can write

$$\begin{aligned} |R[f]| &= \frac{1}{n!} \left| \sum_{i=1}^n \int_{a_i}^{b_i} (x - x_i)^n f^{(n)}(\xi_i) dx \right| \leq \frac{1}{n!} \sum_{i=1}^n \left| \int_{a_i}^{b_i} (x - x_i)^n f^{(n)}(\xi_i) dx \right| \\ &\leq \frac{1}{n!} \sum_{i=1}^n \int_{a_i}^{b_i} |x - x_i|^n |f^{(n)}(\xi_i)| dx \leq \frac{c}{n!} \sum_{i=1}^n \int_{a_i}^{b_i} |x - x_i|^n dx \leq \frac{ch^n}{n!} \sum_{i=1}^n \int_{a_i}^{b_i} dx \leq \frac{ch^{n+1}}{n!}, \end{aligned} \quad (\text{e.10})$$

where $h = |x_n - x_1|$.

Furthermore, the first part of Eq. (e.7) results in

$$\begin{aligned} \sum_{r=1}^{n-1} |\tilde{\mathbf{X}}^{(r)} \mathbf{E}_D^{(r)}| &\leq \sum_{r=1}^{n-1} \frac{1}{2^{r+1}(r+1)!} \\ &\quad \times \left(|x_2 - x_1|^{r+1} |E_D^{(r)}(x_1)| + \sum_{i=2}^{n-1} (|x_{i+1} - x_i|^{r+1} + |x_i - x_{i-1}|^{r+1}) |E_D^{(r)}(x_i)| + |x_n - x_{n-1}|^{r+1} |E_D^{(r)}(x_n)| \right). \end{aligned} \quad (\text{e.11})$$

Using Eq. (e.3), the above relation can be simplified to

$$\sum_{r=1}^{n-1} |\tilde{\mathbf{X}}^{(r)} \mathbf{E}^{(r)}| \leq \sum_{r=1}^{n-1} \frac{ch^{n-r}}{(n-r)!} \frac{2}{2^{r+1}(r+1)!} \sum_{i=2}^n |x_i - x_{i-1}|^{r+1} \leq \sum_{r=1}^{n-1} \frac{(n-1)ch^{n+1}}{(n-r)!2^r(r+1)!}. \quad (\text{e.12})$$

Now, using Eqs (e.7), (e.10), and (e.12), an upper bound of the integration error becomes

$$|E_S(f)| \leq \left(\sum_{r=1}^{n-1} \frac{(n-1)}{(n-r)!2^r(r+1)!} + \frac{1}{n!} \right) ch^{n+1}. \quad (\text{e.13})$$

Having the error equations for the differential and integral operators given by Eqs. (e.3) and (e.13), respectively, we can estimate the error of the VDQ method for a given problem. As an example consider the following functional

$$F(u) = \int_{x_1}^{x_n} \left(\frac{1}{2} \left(\frac{du(x)}{dx} \right)^2 - g(x)u(x) \right) dx. \quad (\text{e.14})$$

Using the VDQ method, we can discretize the above functional as

$$F = \tilde{\mathbf{S}} \left(\frac{1}{2} (\mathbf{D}^{(1)} \mathbf{U} + \mathbf{E}_D^{(1)}) \circ (\mathbf{D}^{(1)} \mathbf{U} + \mathbf{E}_D^{(1)}) - \mathbf{G} \circ \mathbf{U} \right) + E_S, \quad (\text{e.15})$$

where $\circ$ denotes the Hadamard product. By expanding the above equation, we get

$$F = \tilde{\mathbf{S}} \left(\frac{1}{2} (\mathbf{D}^{(1)} \mathbf{U}) \circ (\mathbf{D}^{(1)} \mathbf{U}) - \mathbf{G} \circ \mathbf{U} \right) + \tilde{\mathbf{S}} \left((\mathbf{D}^{(1)} \mathbf{U}) \circ \mathbf{E}_D^{(1)} + \frac{1}{2} \mathbf{E}_D^{(1)} \circ \mathbf{E}_D^{(1)} \right) + E_S. \quad (\text{e.16})$$

Therefore, the approximation error is

$$E[F(u)] = \tilde{\mathbf{S}} \left((\mathbf{D}^{(1)} \mathbf{U}) \circ \mathbf{E}_D^{(1)} + \frac{1}{2} \mathbf{E}_D^{(1)} \circ \mathbf{E}_D^{(1)} \right) + E_S. \quad (\text{e.17})$$

The above relation can be rewritten as follows

$$E[F(u)] = \sum_{i=1}^n \left(S_i U_i^{(1)} E_D^{(1)}(x_i) + \frac{1}{2} S_i (E_D^{(1)}(x_i))^2 \right) + E_S, \quad (\text{e.18})$$

where for $i=1, 2, \dots, n$, S_i and $U_i^{(1)}$ denote the entries of the vectors $\tilde{\mathbf{S}}$ and $(\mathbf{D}^{(1)} \mathbf{U})$, respectively. To find the upper bound of the error, we calculate the absolute value of the above equation as

$$|E[F(u)]| \leq \sum_{i=1}^n |S_i U_i^{(1)}| |E_D^{(1)}(x_i)| + \frac{1}{2} \sum_{i=1}^n |S_i| (E_D^{(1)}(x_i))^2 + |E_S|. \quad (\text{e.19})$$

Assuming that $|u^{(n)}(x)| \leq c_1$ and the n -th order derivative of the integrand of Eq. (e.14) is bounded by the constant c_2 , we can use Eqs. (e.3) and (e.13) to simplify the above equation as

$$|E[F(u)]| \leq w_1 \frac{c_1 h^{n-1}}{(n-1)!} + w_2 \left(\frac{c_1 h^{n-1}}{(n-1)!} \right)^2 + \left(\sum_{r=1}^{n-1} \frac{(n-1)}{(n-r)! 2^r (r+1)!} + \frac{1}{n!} \right) c_2 h^{n+1}, \quad (\text{e.20})$$

$$w_1 = \sum_{i=1}^n |S_i U_i^{(1)}|, \quad w_2 = \frac{1}{2} \sum_{i=1}^n |S_i|. \quad (\text{e.21})$$

The same approach can be used for other problems.

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